



UnionPay Integrated Circuit Card Specifications

— Product Specifications

Part V IC Card Internet Multipurpose Terminal Specification



Version 2014

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Summary of Revisions

The change listed below is associated with the current version.

Description of Change	Where to look
Added: Chapter 8 Certificate Update processing	Chapter 8
Revised: Certificate Verify Message	9.2.1
Added: Channel certificate update description	9.2.3
Revised: Step (6) of handshake protocol operation principle	9.2.3
Revised: P2=0x01 table with more detailed information	A.2.1.4
Revised: Change Table A.17 '1000' value to Terminal Certificate	A.3.2.2
Revised: Update Table A.25 values	A.3.6.4
Revised: Update IC Card Internal Terminal definitions and new name	3.19
Added: Bluetooth interface protocols	11.2
Added: USB OTG protocols	11.3
Added: Appendix H	Appendix H
Added: Appendix I	Appendix I
Added: Appendix J	Appendix J
Added: Appendix K	Appendix K

1 Application Scope

This book applies to all UPI participants.

2 Normative Reference

The clauses in the following documents become the clauses of these Specifications after being quoted by UICS. For the dated references, all their subsequent modifications (excluding contents of the corrigendum) or the revisions are not applicable to these Specifications. However, all the Parties reaching an agreement according to these Specifications are encouraged to study whether the latest version of these documents can be used. For the undated references, their latest versions are applicable to these Specifications.

ISO 9564-1:2002 *Banking -- Personal Identification Number (PIN) management and security -- Part 1: Basic principles and requirements for online PIN handling in ATM and POS systems*

IEC 60950-1999 *Safety of information technology equipment*

CISPR 22-2006 *Information technology equipment— Radio disturbance characteristics— Limits and methods of measurement*

ISO 7810-1985 *Identification cards -- Physical characteristics*

ISO 7811-5-1985 *Identification cards -- Recording technique -- Part 5: Location of read-write magnetic track -- Track 3*

ISO8583-1987 *Bank card originated messages -- Interchange message specifications -- Content for financial transactions*

ISO/IEC 7812-1:1993 *Identification cards -- Identification of issuers -- Part 2: Application and registration procedures*

ISO/IEC 7813:1995 *Identification cards -- Financial transaction cards*

UICS *UnionPay Integrated Circuit Card Specifications*

ISO/IEC 8859 *8-bit single-byte coded graphic character sets*

ISO/IEC 9797-1 *Information technology -- Security techniques -- Message Authentication Codes (MACs) -- Part 1: Mechanisms using a block cipher*
Technical Specifications on Bankcard Interoperability (Version 2.1)

3 Terms and Definitions

The following terms and definitions are applicable to these Specifications:

3.1 Application

The application protocol and relevant datasets shared by card and terminal.

3.2 Asymmetric Encryption Technique

The usage of two interrelated encryption transformation techniques: public transformation (defined by public keys) and private transformation (defined by private keys). These two types of transformations are unique in that when undergoing public transformation a private transformation cannot be calculated.

3.3 Authentication

The process by which an entity is confirmed to have the identity that it claims to have.

3.4 Card

Card in these Specifications refers to consumer devices with contact and contactless payment application and interaction function with payment terminal.

3.5 Certificate

A signature process applied to entity's public key, identifying information and other relevant information by the authentication centre which generates certificate using private keys, thus producing unforgeable data.

3.6 Certificate Authority

A trustworthy third party institution that authenticates the public keys and other relevant information of a given entity; this term is simplified in these specifications are "CA Authentication Center" or "CA Center"; unless otherwise stated, the term "CA Center" in these specifications refer to UnionPay while "CA Center Server" refers to the financial IC card internet terminal certificate management system, simplified as "RA System".

3.7 Certification Authority Root Certificate

The starting point of a chain of trust whereby the CA authentication center gives itself an unsigned public key certificate or self-signing certificate.

3.8 Command

A message sent from a terminal to an IC card that either initiates a process or requests a response.

3.9 Cryptogram

The result of an encryption operation.

3.10 Encryption Algorithm

A transformation algorithm that either hides or displays the informational content of data.

3.11 Decryption

The reverse operation of an encryption.

3.12 Digital Signature

A type of asymmetric encryption transformation performed on data. This transformation allows for the receiver of data to confirm the source and integrity of the data, protects the sender and receiver of the data from being modified by a third-party, and also protects against data being modified by the receiver.

3.13 Distinguished Name

A uniquely identifying certificate user name for digital certificates.

3.14 Encryption

A process by which an encryption algorithm is used to perform a reversible transformation of data to create a ciphertext.

3.15 Financial IC Card

Integrated circuit cards issued by merchant banks in compliance with UICS requirements.

3.16 Handshake Protocol

The processes used before an actual transmission of data whereby the identity of all parties are authenticated, encryption algorithms are determined, encryption keys are exchanged, etc.

3.17 Integrated Circuits

Electronic devices capable of processing and storing information.

3.18 Integrated Circuit(s) Card (ICC)

A card that is packed internally with one or more ICs used for processing and storage.

3.19 IC Card Internet Terminal

A small card reading device that utilized an internet channel to complete an IC card transaction together with an IC card; the device includes interface devices and all relevant components and interfaces (such as computer interfaces, etc.). [Based on

this specification, all internet terminal products developed by UnionPay are named as 'MiniPay'.

3.20 Interface Device

The mechanical and electronic components of a terminal for accepting IC input.

3.21 Management Command

Operational commands used by terminals for receiving terminal parameter data and controlling terminal prompts.

3.22 Pin Encryption Certificate

A digital certificate issued by Root CA Center used to encrypt online PIN digital certificates in X.509 format.

3.23 Private Key

An asymmetric key used by an entity on data containing identifying entity information; also used to apply digital signatures.

3.24 Processing Center

A system used to receive, process, and transmit terminal transaction request information as well the transmission of transaction result information back to terminals; within the standard, all references to processing centers refer to the UnionPay Micropayment Service Processing System.

3.25 Public Key

One of the key of an entity's asymmetrical key pair which could be disclosed. Used for authentication of digital signatures.

3.26 Public Key Certificate

Unforgeable entity public key information signed by the authentication authority.

3.27 Load

The process of increasing an electronic cash balance. There are multiple load methods; transferring from a primary account, cash deposit, or fund transfers from other accounts. However the final post-loading balance could not exceed the electronic cash limit.

3.28 Record Protocol

Basic functionality based on a dependable transmission protocol that support data packaging, compression, and encryption for application layer protocols.

3.29 Response

The message sent to a terminal by an IC card signalling that it has processed a received command message.

3.30 Response Code

Also called an answer code; a code sent by a receiver of a request or notification that indicates to the sender the result of how that request or notification was processed.

3.31 Secure Channel

A secure communication channel formed between an IC internet terminal and processing center.

3.32 Script

A command or command sequence sent from an issuer, used for inputting that command into an IC card.

3.33 Secure Channel Command

A command used by a terminal to construct a secure channel with a processing center and manage electronic signatures.

3.34 Symmetric Encryption Technique

An encryption technique whereby the sender and receiver both use the same encryption key for data transformation; without this key it is impossible to derive the data transformation conducted either by the sender or receiver.

3.35 Terminal Certificate

A unique digital certificate compliant with X.509 formatting used to identify a terminal device; each terminal device will have this certificate written into it during pre-personalization.

3.36 Transaction Command

The commands used by terminals to execute transaction initialization and online processing of debit/credit application procedures.

3.37 Trusted server certificate Trusted server certificate

A unique digital certificate compliant with X.509 formatting used to identify processing center system servers; each processing center server has its own unique trusted server certificate.

3.38 Trusted platform module

Trusted platform module is a device that can perform key generation, encryption and decryption independently with independent internal processor and memory cell,

and can store keys and the featured data, providing encryption and security certification service for the equipment. It uses security chip for encryption and the key is stored in the hardware so that the stolen data cannot be decrypted, thus protecting commercial privacy and data security.

3.39 UUID

Universally Unique Identifier.

4 Symbols and Abbreviations

The following table of symbols and abbreviations is applicable to these Specifications:

AID	Application Identifier
an	Alphanumeric
APDU	Application Protocol Data Unit
ATC	Application Transaction Counter
BCD	Binary Coded Decimal
CA	Certificate Authority
CBC	Cipher Block Chaining
CLA	Class Byte of the Command Message
CCID	USB Chip/Smart Card Interface Devices-USB
DES	Data Encryption Standard
DN	Distinguished Name
DOL	Data Object List
FIPS	Federal Information Processing Standard
HMAC	Keyed-Hash Message Authentication Code
INS	Instruction Byte of Command Message
LED	Light Emitting Diode
MAC	Message Authentication Code
n	Numeric
P1	Parameter 1

P2	Parameter 2
PAN	Primary Account Number
PIN	Personal Identification Number
RA	Register Authority
RSA	Rivest, Sharmir and Adleman Asymmetric Key Algorithm
SHA-1	Secure Hash Algorithm
Terminal	Terminal in these Specifications refers to UICS IC card internet terminal
USB	Universal Serial BUS
X.509	Digital Certificate Standard defined by the Telecommunication Standardization Bureau of the International Telecommunication Union

5 Requirements for Terminal Hardware Specifications

5.1 Requirements for Terminal Security

Financial IC card Internet terminal adopts the hardware encryption module to ensure the secure input and encrypted processing of sensitive information, such as personal identification code (PIN), support the establishment of security channel with the processing center, and perform the encryption and decryption operations for data interacting with external entities, and the validity and integrity verification. Terminal could store keys in security, prohibit direct access and output of keys, and prevent illegal injection, replacement and use of keys through effective security mechanism. It shall ensure that the firmware and software of terminals will not be illegally injected or updated.

Terminal shall store and transmit sensitive information, such as bankcard number, card verification code, PIN and card validity period, in ciphertext.

Only asterisks should be displayed and not plaintext whenever a cardholder inputs a password into a terminal.

When a cardholder is asked to input the PIN, the terminal should not allow the cardholder to skip this process; in the other words, PIN bypass is not allowed; only after the cardholder has inputted the six digits PIN, should the terminal allow for the continuation of the transaction process.

At the very least, the secure memory space of a terminal should satisfy the requirement for storing transaction certificates and keys that are discussed in other sections of these specifications.

Terminals should only response to the instruction requests that are contained within the scope and definitions defined by the command instruction set of these specifications; all instructions outside of this instruction set should be ignored.

5.2 Secure Module Requirements

Terminals should use secure modules that are capable of performing both key generation and digital signature calculations, guarantee that sensitive operations are only performed within these secure modules, and guard against both the leakage of sensitive information and detrimental influences to security functions.

Secure modules should contain unreadable areas that are used to store unique terminal information such as terminal private keys. The following mechanisms are not allowed: the capability to output private key or PIN plaintext or the ability to use leaked keys to encrypt PINs or other keys.

The random number used during key calculations should be generated within the secure module; the randomness of this number should be in accordance with the

requirements set by international general hardware-generated random number standards.

5.3 Hardware Composition

5.3.1 IC Card Reader Module

Terminals should be equipped with an IC card reader module; this module should be capable of communicating command data with IC cards and support both contact and contactless IC cards. This module should include mechanical, electrical, and logical protocol components; for specific requirements please see UICS.

Terminals should be equipped with markings that illustrate how to insert contact IC cards or read contactless IC cards.

5.3.2 Display Screen

Terminals should be reequipped with display screens that allow for the monitoring of transaction process, input of data, and the setup, selection, and confirmation of transaction data. Terminals should support ISO8859 basic character sets. The display screen should be capable of displaying Chinese, English, and numbers.

5.3.3 Keyboard

Terminals should be equipped with button keyboards that allow for the input of transaction amounts, PIN numbers, command selections, and execution functionality. For digital keyboard, letter button, command button, function button, and button layout requirements, see UICS. If colored command buttons are used, the following colors are recommended.

Command button colors: Enter – green; cancel – red; clear – yellow.

Financial IC card internet terminal keyboards should have at least 10 number buttons, and an appropriate number of function buttons; function buttons must include cancel, clear, and enter as well as up and down function buttons. Keyboard output passwords should be displayed but only in a meaningless form that represent the inputted digits of the password.

5.3.4 Communication with Upper Computer

Communication port shall at least support USB communication methods:

5.4 Power Supply

Terminal shall at least support power supply in the form of USB. According to the existing requirements for USB power supply, power supply voltage should be DC $5V \pm 5\%$, and current should be less than 500mA; if having built-in battery, terminal shall meet the relevant certification requirements imposed by China's compulsory product certification system.

5.5 Electromagnetic Compatibility

Radio interference limit shall comply with A level ITE stipulations in CISPR 22-2006.

Electromagnetic sensitivity shall comply with the stipulations in HP 765 -1977.

5.6 Reliability

Unless otherwise specified for special parts, the consecutive operation time between failures shall be no less than 50,000 hours.

5.7 Terminal Types

After connecting to a computer using a USB interface, financial IC card internet terminals typically enumerates the name of the terminal and then uses this terminal name to identify the terminal type. The terminal name rules are as follows:

‘CUP_R’+ terminal type + ‘_’ + vendor code + ‘_’ + vendor type number

Table 1 Terminal Type Table

Terminal Type	Meaning
1	General Contact Reader
2	LCD and Keyboard-Equipped Contact Reader
3	General Dual Interface Reader
4	LCD and Keyboard-Equipped Dual Interface Contact Reader
5	General Contactless Reader
6	LCD and Keyboard-Equipped Contactless Reader

Notes: 1. ‘_’ indicates a blank space;

2. Merchant code and merchant type number must be a string consisting of English letters and numbers; the total length of the merchant type number must be smaller than 20 bytes.

For hardware requirements for various types of terminal, please refer to Table 2.

Table 2 Hardware Requirements for Terminal Types

S/N	Hardware Types Hardware module	Contact financial IC card internet terminal	Contactless financial IC card internet terminal	Dual interface IC card internet terminal
1	Keyboard	Mandatory	Mandatory	Mandatory
2	Display Screen	Mandatory	Mandatory	Mandatory
3	Contact IC card reader module	Mandatory	None	Mandatory
4	Contactless IC card reader module	None	Mandatory	Mandatory
5	Host Communication module	Mandatory	Mandatory	Mandatory
6	Security module	Mandatory	Mandatory	Mandatory

6 General requirements

6.1 Transaction type

Financial IC card internet terminal shall at least support transactions such as E-cash load, electronic cash balance inquiry and debit/credit primary account balance inquiry.

6.2 Supported Card Mediums

Terminals should support the reading and writing of financial IC cards and industry IC cards but not support magstrip cards.

6.3 Download Management

Terminal shall be able to provide the secure download, update and deletion of certificate, terminal program and parameters.

Download method shall be online remote download. Terminal shall ensure the security of download control. Only the authorized or approved party could download data to terminal, and it is not allowed to modify contents in the terminal without authorization. Terminal shall also be able to confirm the security of downloaded data, verify the integrity and correctness of terminal download program and ensure that the sensitive and crucial key data will not be disclosed in the download process.

For terminal firmware that has been released to users, the manufacturers could determine at its sole discretion whether to update the terminal firmware through the method of automatic download by terminals. The format of the firmware could be determined by the manufacturers themselves as well.

7 Terminal Personalization

7.1 Terminal Personalization Data

Terminal personalization data shall at least include terminal data, terminal certificate, Root CA certificate and PIN encryption certificate.

7.1.1 Terminal Data

Terminal data is comprised of the following information: affiliated institution code, affiliated institution self-defined data (including merchant identifier code and terminal manufacture date), and terminal identifier code. Terminal data should be preset before it leaves its manufacturing facility and must not be modified afterwards.

Affiliated Institution Code: this code consists of 8 digits and consists of bank code + area code. If the 4-digit area code is not defined then it can be replaced with 0000; for instance, 01022900 represents the Shanghai branch of ICBC, 01045800 represents the Guangdong branch of BOC, 00010000 represents UnionPay, etc.

Merchant Identifier Code: this code consists of 3 numbers and is distributed by UnionPay to each merchant.

Terminal Manufacture Date: this code consists of 4 numbers with a format of YYMM; for instance, a manufacture data of August 2011 will be represented as 1108;

Terminal Identifier Code: this code consists of 8 bytes of numbers or letters; terminal identifier codes are created when a terminal's affiliated institution applies for them from UnionPay, which distributes them uniformly. Affiliated institutions then either hand them over to the terminal vendor they have for terminal personalization, or UnionPay itself can directly hand over the terminal identifier code to the terminal vendor chosen by the affiliated institution for personalization.

Table 3 Terminal Data

Explanation	Length (in bytes)	Type
Affiliated Institution Code	8	n
Merchant Identifier Code	3	n
Terminal Manufacture Date	4	n
Terminal Identifier Code	8	ans

7.1.2 Terminal Certificate

Terminal certificates must be attained through an application to UnionPay by a terminal's affiliated institution and are issued by UnionPay using the RA system as X.509 format public key certificate; see Appendix D – Certificate Format Table for format content details. This certificate is used to identify a terminal's legal and unique public key certificate, the public key of which is generated by the terminal.

7.1.3 Root CA Certificate

Root CA certificates are issued and managed by UnionPay using the RA system, where they must also be downloaded. They are primarily used to verify terminal certificates, trusted server certificates, and the legality of PIN encryption certificates throughout the transaction process.

7.1.4 PIN Encryption Certificate

PIN certificates are attained by an application submitted to and downloaded from the CA center by the processing center; they are primarily used for online PIN encryption. For two UICS format public keys of different lengths (1024 and 2048 bits), two certificates will be generated by a UnionPay encryption device (HSM). For a terminal contains more than one PIN encryption certificate then the terminal will choose the PIN encryption certificate to encrypt the online PIN according to the CN value of the trusted server certificate's DN field. For DN field rules and double processing center processing requirements see Appendix F.

7.2 Terminal Public Key

Terminal public keys are generated by the terminal before the terminal download of the terminal certificate. Terminal generated public keys are submitted to CA for creation of certificates; terminal private keys should be stored within the terminal secure module and not be exported at any time.

7.3 Terminal Personalization Procedures

Terminal personalization occurs before a terminal leaves its manufacturing facility and consists of terminal personalization data being written into the terminal; of this data, the Root CA certificate, terminal certificate, and PIN encryption certificate must be downloaded from CA Center servers.

Specific procedures are as follows:

- Terminal data is written into the terminal;
- Terminal certificates are installed into the terminal device;
- Root CA certificates are installed into the terminal device;
- PIN encryption certificates are installed into the terminal device.

7.4 Certificate Application and Issuance Procedures

Certificates associated with financial IC card internet terminals include trusted server certificates, terminal certificates, Root CA certificates, and PIN encryption certificates.

7.4.1 Root CA Certificate

Root CA certificates are used to verify the authenticity and legality of trusted server certificates, terminal certificates and PIN encryption certificates and they must be written into the secure communication devices of financial IC card internet devices and processing centers during the personalization process. Root CA certificate downloads are negotiated and set by UnionPay and terminal affiliated institutions together.

7.4.2 Terminal Certificate

Before applying for a certificate, terminals must conduct a security appraisal of the certificate application channel; only if this appraisal is successful should the terminal initiate certificate application and issuance processes. Once CA Center appraisal is successful, application and issuance procedures proceed as shown in the image below.

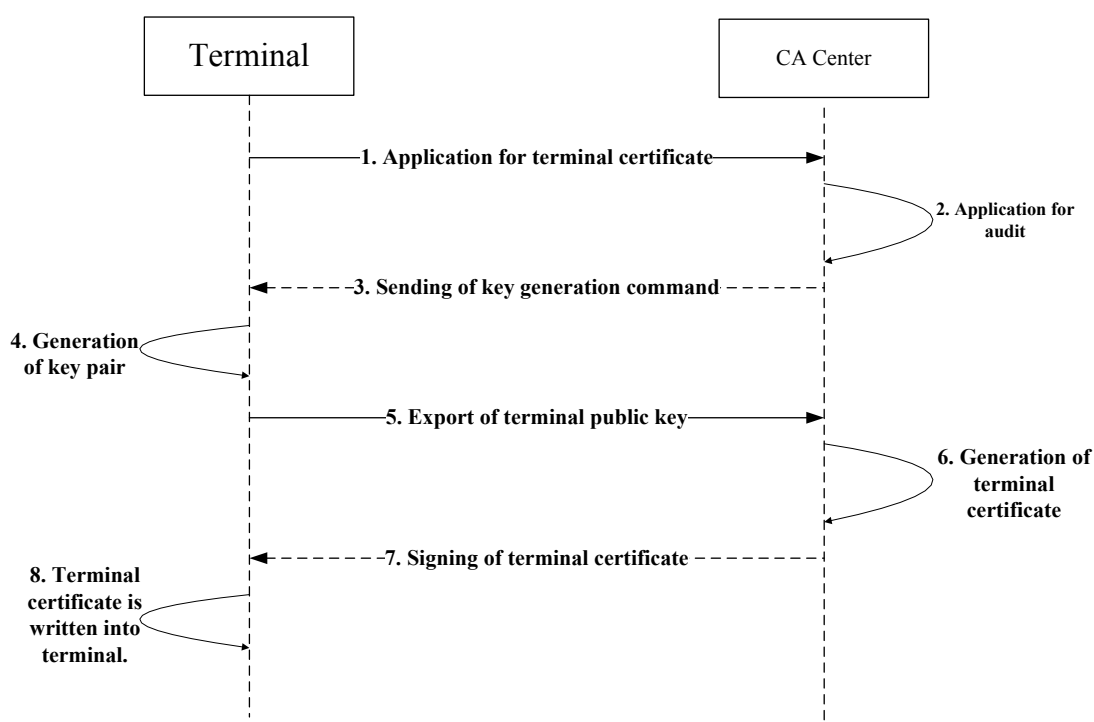


Chart 1 Terminal Certificate Issuance Procedures

Specific procedures are as follows:

- 1) Terminal submits terminal certificate application;

- 2) CA Center audits the terminal certificate application;
- 3) CA Center issues terminal key pair generation command;
- 4) Terminal generates public/private key pair;
- 5) Terminal exports terminal public key;
- 6) CA Center generates terminal certificate;
- 7) CA Center signs terminal certificate;
- 8) Terminal certificate is written into terminal.

The terminal vendors shall develop the download tools according to the interface defined by CA center; terminal certificates can only be downloaded from the RA system directly into internal storage of a terminal using these download tools; no other download method is permitted and only terminals may conduct such downloads.

7.4.3 Trusted server certificates

Trusted server certificates can be used to verify the authenticity of processing center servers, guard against server fraud, and server authentication during secure communication with terminal devices. Certificates are standard X.509 certificates; see Appendix D – Certificate Formats for format content information. Trusted server certificates are preset in processing center servers, which each receive a unique server certificate. The issuance processes for these certificates is the same as for terminal certificates and are only appropriate for self-constructed processing center institutions.

7.4.4 PIN Encryption Certificate

Used by financial IC card internet terminals to protect financial transaction PINs during transactions. Financial IC card internet terminals must be preset with PIN encryption certificates before they leave the factory, and the application and issuance processes are the same as those for terminal certificates; for dual processing centers or multi-processing center modes, UnionPay PIN encryption certificates should be set first and PIN encryption certificates of other institutions be set later.

8 Certificate Update

8.1 Certificate Files

In the daily usage of terminal, it has to support the update of certificates: CA root certificate, Channel certificate and PIN encryption certificate.

8.2 Certificate Update Flow

8.2.1 CA Root Certificate Updates

The update of CA root certificate and Terminal certificate happen at the same time. During the update of terminal certificate, backend system decides if CA certificate file needs to be updated based on the CA certificate serial number returned by terminal, and distributes new CA root certificate before distributing new terminal certificate. The detailed update flow refers to Appendix I.1.

8.2.2 PIN Encryption Certificate Update

The update PIN encryption certificate depends on MiniPay transaction system. MiniPay transaction system decides if PIN encryption certificate file needs to be updated based on certificate serial number in the terminal. The whole update process will be performed under secure channel. The detailed update flow refers to Appendix I.2.

8.2.3 Terminal Certificate File update

After secure channel is established, backend system requests terminal to generate P10 certificate request file based on the validation date of terminal certification returned by terminal, connects RA system to distribute new certificate, and sends terminal certificate to terminal device. The detailed update flow refers to Appendix I.3.

9 Security System

9.1 Certification System

9.1.1 CA System Structure

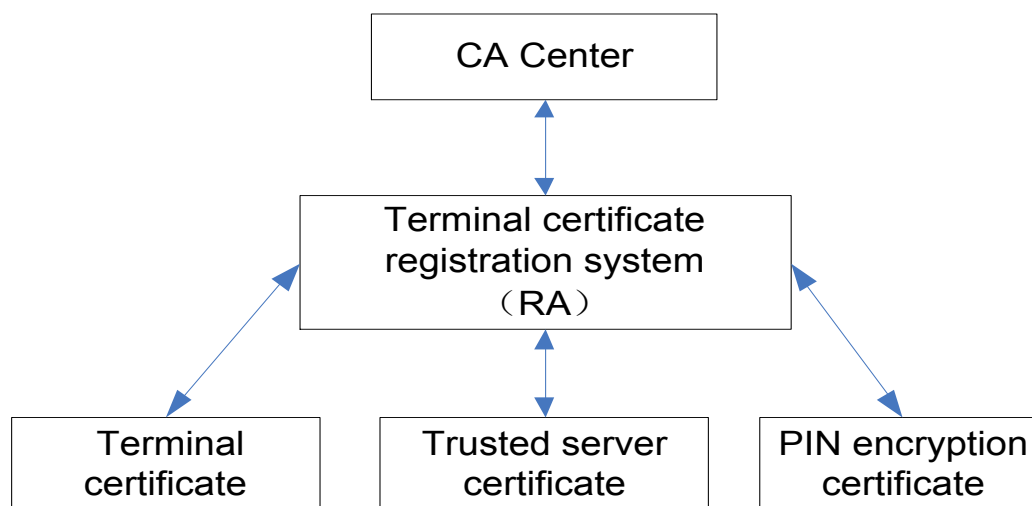


Chart 2 Terminal CA System Structure

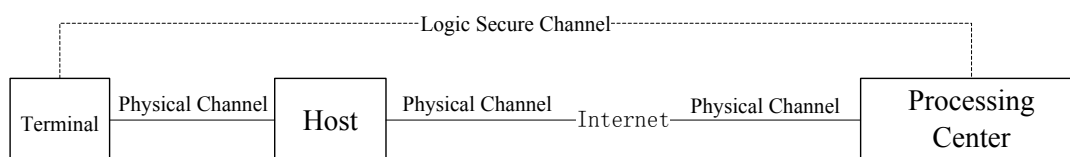
As shown in the image above, the terminal certificate registration system (RA) is primarily used to audit applications from terminal merchants and processing centers and then issue certificates to terminals and processing centers once auditing is completed.

9.1.2 Private Key Algorithm

Key Algorithms use UICS approved symmetrical, asymmetrical, and hash algorithms with simultaneous key length support of 1024 and 2048 bits; the RSA algorithm and signature algorithm used for the generation of certificate public and private keys within these specifications use SHA-1 and support SHA256 expansion.

9.2 Secure Channel

Financial IC card Internet terminal accesses to processing center through the networking devices (such as PC) that it connects to, and establishes the terminal-to-terminal logic secure channel with the processing center through the networking device adopting handshake mechanism. Its overall framework is shown in Chart 3

**Chart 3 Secure Channel Chart**

The protocol establishment of secure channel is composed of two parts: Handshake Protocol and Record Layer Protocol. Handshake Protocol is used to complete the two-way identification between terminal and server and the exchange process of session key. Record Layer Protocol is used to complete the encrypted transmission of application data.

9.2.1 Message Code

The basic data unit in protocol interaction process is called message. Message includes three parts of message type, message length and message body. In these Specifications, the definition of message code shall be described in C language format. Basic message unit is defined as follows:

```

struct {
    MessageType msg_type;
    WORD      length;
    BYTE      body[ ];
} Message;
  
```

Wherein, message type `msg_type` is a 1 byte value of enumeration type, message length (`length`) is a 2 bytes nonnegative integer, and message body is a variable-length data structure and shall be defined according to different message type. Message type is defined as follows:

```

enum{
    client_hello = 0x80,
    server_hello = 0x81,
    server_certificate = 0x82,
    client_certificate = 0x83,
    certificate_verify = 0x84,
    client_key_exchange = 0x85,
    server_finished = 0x86,
    client_finished = 0x87,
  
```

```

        error_message= 0x88,

        application_data = 0x89

    } MessageType;

```

Message can be divided into three types according to different functions of message in protocol interaction process, i.e., handshake protocol message, error information message and record protocol message.

(1) Handshake Protocol Message

Handshake protocol message is used to complete the two-way identification between terminal and Internet server and the exchange process of session key, defined as follows:

```

struct {

    MessageType msg_type;

    WORD        length;

    switch ( MessageType ) {

        case client_hello :      ClientHello;

        case server_hello:      ServerHello;

        case server_certificate: ServerCertificate;

        case client_certificate: ClientCertificate;

        case certificate_verify: CertificateVerify;

        case client_key_exchange: ClientKeyExchange;

        case server_finished:   ServerFinished ;

        case client_finished:   ClientFinished;

    } body;

    } Message;

```

- ClientHello Message

32-byte random number generated by terminal is defined as follows; for cryptographic algorithm, please refer to Table 3 cipherSuite.

```

struct {

    BYTE random[32];

    BYTE cipherSuite[2];

    } ClientHello;

```

The first byte is used in the Specifications. The second byte shall be reserved. For the definition of the first byte of cipherSuite, please refer to Table 4.

Table 4 CipherSuite

B ₇	B ₆	B ₅	B ₄	B ₃	B ₂	B ₁	B ₀	Algorithm
X	X	X	X	*	*	X	1	RSA
X	X	X	X	*	*	1	X	ECC
X	X	X	1	*	*	X	X	3DES
X	X	1	X	*	*	X	X	AES-128
X	1	X	X	*	*	X	X	3DES
1	X	X	X	*	*	X	X	AES-128

Note: B2 and B3 bit in cipherSuite byte shall be reserved. Corresponding position in ClientHello being 1 represents the cryptographic algorithm set the terminal supports. Similarly for ServerHello, corresponding position being 1 represents the cryptographic algorithm selected by the server terminal. ServerHello message is defined as follows:

- ServerHello Message

32-byte random number generated by server terminal is defined as follows; for cryptographic algorithm, please refer to Table 4 cipherSuite.

```
struct {
    BYTE random[32];
    BYTE cipherSuite;
} ServerHello;
```

- ClientCertificate Message

X.509 certificate of terminal is defined as follows. Including RSA public key of client, used to verify the CertificateVerify message.

```
struct {
    BYTE certificate[ ];
} ClientCertificate;
```

- ServerCertificate Message

X.509 identification certificate of server terminal is defined as follows, including RSA public key of server, and the client uses such public key to encrypt the sharing secrets.

```
struct {
    BYTE certificate[ ];
} ServerCertificate;
```

- CertificateVerify Message

The Client adopts RSA private key to sign the results of linking ClientHello and ServerHello messages. For signature algorithm, please refer to Appendix B.2. Signature result: $\text{signature} = \text{Sign}(\text{Master_secret} || \text{ClientHello} || \text{ServerHello})$ as message main body.

```
struct {
    BYTE signature[ ];
} CertificateVerify;
```

- ClientKeyExchange Message

It is 48-byte master_secret that uses server terminal certificate public key for encryption. For cryptographic algorithm, please refer to Appendix B.1.

```
struct {
    BYTE encryptedSharedSecret[ ];
} ClientKeyExchange;
```

- ServerFinished Message

Verification message of server terminal for handshake process is defined as follows:

$$\text{message_MAC} = \text{HMAC}(\text{master_secret}, \text{Finish_label} || \text{Hash}(\text{handshake_messages}))$$

For HMAC algorithm, please refer to Appendix C.2, among which, master_secret is the master secret, Finish_label is the 6-byte ASCII code value “SERVER”, and Hash algorithm adopts the SHA-1. Handshake_messages is the link of handshake message:

```
handshake_messages =
(ClientHello||ServerHello||
Hash(ServerCertificate)||
Hash(ClientCertificate)||
```

CertificateVerify||ClientKeyExchange);

struct {

 BYTE message_MAC[32];

} ServerFinished;

- ClientFinished Message

Verification message of client for handshake process is defined as follows:

$$\text{message_MAC} = \text{HMAC}(\text{master_secret}, \text{Finish_label} || \text{Hash}(\text{handshake_messages}))$$

For HMAC algorithm, please refer to Appendix C.2, among which, master_secret is the master secret, Finish_label is the 6-byte ASCII code value “CLIENT”, and Hash algorithm adopts the SHA-1. Handshake_messages is the link of handshake message:

handshake_messages =

(ClientHello||ServerHello||

Hash(ServerCertificate)||

Hash(ClientCertificate)||

CertificateVerify||ClientKeyExchange);

struct {

 BYTE message_MAC [32];

} ClientFinished;

(2) Error information message

Error information message is used for the error processing in protocol interactive process. It is defined as follows:

struct {

 MessageType msg_type = error_message;

 WORD length;

 BYTE body;

} Message;

Wherein, for the definition of message type and message body, please refer to Table 5:

Table 5 Definition of Error Information Message

Name of message body	Message body code	Definitions of message body
ClientCiperSuiteError	01	Sent by server, indicating the terminal password algorithm cipherSuite error.
ServerCiperSuiteError	02	Sent by terminal, indicating server terminal password algorithm cipherSuite error.
ClientCertificateError	03	Sent by server, indicating terminal certificate verification error.
ServerCertificateError	04	Sent by terminal, indicating server terminal certificate verification error.
ClientHandshakeError	05	Sent by server, indicating handshake authentication sent by terminal error.
ServerHandshakeError	08	Sent by terminal, indicating handshake authentication sent by server terminal error.
RecordError	09	Sent by server or terminal, indicating record layer protocol data transmission error.

(3) Record layer protocol message

Handshake protocol will enter record layer protocol after completing the key agreement. Record layer message is used for application data transmission, defined as follows:

```

struct{
    MessageType  msg_type = application_data;
    WORD         length;
    BYTE         encryptedData [ ];
    BYTE         dataMac[8];
} Record;
```

Wherein, encryptedData is the application data transmitted in secure channel after encryption. For data cryptographic algorithm, please refer to Appendix B.1. dataMac is the 8-byte message authentication code of data. For message authentication code algorithm, please refer to Appendix B.2. Length is the total length of encryptedData and dataMac.

9.2.2 Channel Certificate update

Channel certificate update is configured and changed by backend system. To meet the requirements of different terminal configurations, backend system needs to support different certificate types with different encryption key length. Configuration flow and related detailed requirements refer to Appendix I.3.

9.2.3 Handshake Protocol Operation Principle

Handshake protocol is used to complete mutual identification between terminal and server and the exchange process of session key.

Basic procedure of handshake protocol

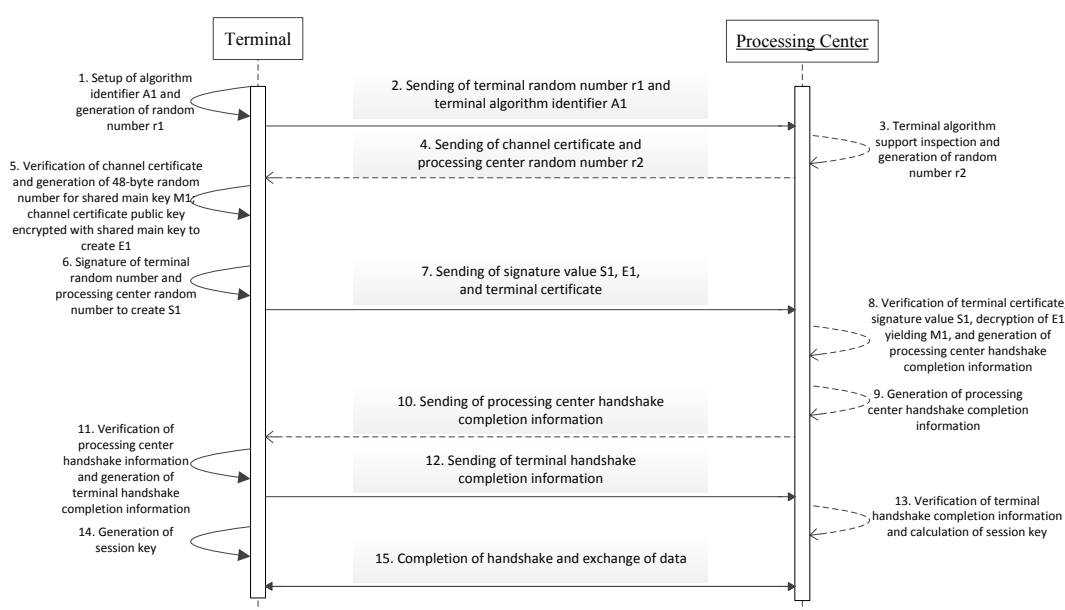


Chart 4 Handshake Protocol Message Sequence Diagram

Handshake protocol working steps

- 1) Terminal acquires algorithm identifier A1 and produces random number r1; r1 and A1 are combined to produce R1 ($R1=r1|A1$), depending on the algorithm support settings of the terminal, the following steps will require the usage of either symmetrical or asymmetrical algorithms;
- 2) Terminal sends random number and algorithm information to the processing center, thus initiating the handshake protocol;
- 3) The processing center selects algorithm identifier A2 and produces random number r2; r2 and A2 are combined to create R2. The processing center checks if it can support the algorithm information provided by terminal; if the processing center supports the algorithm then it will then initiate the

- encryption algorithm; if not supported the processing center will return an error message and break the connection;
- 4) The processing center transmits a random number and processing center's trusted server certificate;
 - 5) The terminal uses a preset Root CA certificate to verify the authenticity of the received processing center trusted server certificate; if the verification is not successful then an error message will be generated and the connection will be stopped; otherwise, the terminal will generate a 48-byte random number to act as a shared main key M1, the card will then use the asymmetrical algorithm established before using the public key provided by processing center trusted server certificate to encrypt M1 and generate E1;
 - 6) M1 connects with R1 and R2, then create R3; the terminal will use digest algorithm on R3 to create H1, then use its own private key to sign H1, and creating S1;
 - 7) The terminal sends S1, E1, and terminal certificate to the processing center;
 - 8) The processing center uses the Root CA certificate to verify the legality of the terminal certificate; If the verification fails then an error message will be produced and the connection will be stopped; if verification succeeds then the terminal certificate will be used to verify S1. If S1 verification fails then then an error message will produced and the connection will be stopped; otherwise, E1 will be decrypted to yield shared main key M1;
 - 9) The processing center will conduct a summary operation on the trusted server certificate to obtain H2 and on the terminal certificate to obtain H3. R1, R2, H2, H3, S1, and E1 are connected to obtain T1 ($T1=R1||R2||H2||H3||S1||E1$); a digest operation will then be conducted on T1 to obtain H4; ASCII code "SERVER" and H4 are connected to yield D1; the first 16 bytes of M1 are used to conduct HMAC on D1 to obtain F1 (See Appendix C.2 for HMAC calculation algorithms);
 - 10) The processing center sends verification of handshake completion F1 to the terminal;
 - 11) The terminal will verify F1 sent from the processing center; if verification is not successful then then an error message will be produced and the connection will be stopped; if verification succeeds then the terminal will produce handshake verification message F2; F2 calculation is performed differently than that used for F1 and only requires that the ASCII code "SERVER" obtained during F1 to be changed to ASCII code "CLIENT";
 - 12) The terminal sends verification of handshake completion F2 to the processing center;

- 13) The processing center will use the same calculation process to verify F2. if verification is not successful then an error message will be produced and the connection will be stopped;
- 14) Once the handshake procedure above is successful then both sites will use the following method to calculate a session key:

$$X = \text{HMAC}(M1, \text{key_label} \parallel R1 \parallel R2) \quad (\text{M1 uses its first 16 bytes})$$

Key_label is the 3-byte ASCII code “key” ;see Appendix B.2 for HMAC algorithm. X1X2...X20 are the first twenty bytes of X and encryption key SKey is: SKey = X1X2...X16; MAC key MKey is: MKey = X5X6...X20;

- 15) Handshake completed.

9.2.4 Record Layer Protocol Working Principles

After successful handshake, both parties could conduct the data transmission in the established secure channel

Data encryption method of record layer protocol

Add data block length (2-byte) before the transmitted data (Data) to form data block $D = (\text{Length} \parallel \text{Data})$. Use encryption key Skey to encrypt D according to cryptographic algorithm specified by the processing center and the terminal. Namely:

$$EData = E_{SKey}(D);$$

Method for Protection of Data Integrity of record layer protocol

In the transmission process of record layer protocol, designate a record serial number for each sending and receiving record at both sender and receiver. Its initial value Seq0 shall be generated as:

The first 8 bytes of multi-interface card random number Random1 are acquired as well as the first 8 bytes of processing center random number Random2, thus $\text{Seq0} = \text{Random1} \parallel \text{Random2}$.

For each record of a sent or received message, the record serial number will add a 1, namely $\text{Seq}_i = \text{Seq}_{i-1} + 1$. Note that all parties must maintain serial number synchronization.

The integrity of application data from both parties should be protected using information authentication code MAC; MAC is generated using the following method:

$$\text{DataMAC} = \text{MAC}(\text{MKey}, \text{Seq}_i \parallel EData) \quad (\text{MKey uses its front 16 bytes})$$

EData is the transmitted encrypted application data; Seq_i is the current record serial number. See Appendix C.1 for MAC calculation method. Once the terminal or

processing center has received data it will then verify MAC authenticity; if verified successfully then processing will continue; if not, an error message will be generated and the connection will be terminated..

10 Terminal Transaction Flow

This section describes online transaction procedures for financial IC card internet terminals; these transaction processing procedures occur after the terminal and processing center have authenticated each other and created a secure channel.

10.1 Transaction Initialization

Once the terminal and processing center have established a secure channel, the terminal will begin to analyze the transaction based on the commands received from the processing center (see Appendix A.4 for command details); if there are no processing errors then the debit/credit transaction processing procedures will be initiated.

For load transactions, the terminal should automatically check electronic cash upper balance limits and inform the cardholder of the maximum allotted load balance.

10.2 Application Selection

Refer to UICS for description.

10.3 Application Initialization/ Reading of Application Data

Refer to UICS for description.

10.4 Offline Data Authentication

MiniPay only supports online transactions; terminals do not support offline data authentication.

10.5 Processing Limits

See UICS for description.

10.6 Cardholder Authentication

When MiniPay is undergoing online transaction procedures it must enact online PIN verification. Terminals should display transaction balances and indicate the cardholder input online PIN on the terminal.

10.7 Terminal Risk Management

Terminal risk management setup requires this transaction to be conducted online; see UICS for a description of other risk management methods..

10.8 Terminal Behavior Analysis

See UICS for description.

10.9 Card Behavior Analysis

Refer to UICS for description.

10.10 Online Processing

Refer to UICS for description.

10.11 Transaction Conclusion

Refer to UICS for a detailed description. Terminals should use light and sound to notify cardholders that a transaction is completed.

10.12 Issuer Script Processing

Regarding successful online issuer transactions, should an authorization response message contain scripts then the terminal must derive script commands from the script and send them to the IC card for execution.

For load transactions, should a terminal receive an explicit script failure or issuer authentication failure then it should initiate a reversal; otherwise terminals should not initiate reversal procedures, including for IC card issuer script response time-out situations.

11 Terminal Interface Protocols

11.1 USB Interface Protocols

Terminal uses the USB interface and adopts CCID protocol. For the the command format sent to terminal by the Host, please refer to the following Table:

Table 6 Command Format Sent to Terminal by the Host

Information field	Identifier	Byte length	Meanings
Communication data head	Type	1	CCID command
	Length	4	Length of Abdata
	Slot	1	Card slot number
	Bseq	1	Result number
	bBwi	1	Block waiting time
	Level Param	2	Selection of communication mode
Command	Abdata		Data sent to terminal

Example of the random number command read by financial IC card:

6F	05000000	00	F1	00	0000	0084000008
↓	↓	↓	↓	↓	↓	↓
Type	Length	Slot	Bseq	bBwi	Level	Abdata

11.2 Bluetooth Interface Protocol

11.2.1 Service Compliance

Bluetooth communication requires terminal to support 3.0 or 4.0 protocol, Bluetooth 3.0 does not have service compliance, and Bluetooth 4.0 reads and writes service need to follow below compliance:

Table 7 Bluetooth 4.0 Service Compliance Code

Service Name	Code(UUID)
Read Service	b80ef097-5332-482a-a0bb-795402319a2f
Write Service	1193439d-c7cf-4494-83ce-7dc7079c7b3f
Read characteristic	5712bd0c-6ffd-4890-ad1e-90a4c0f9ceec
Write characteristic	25ccb710-151b-4d8e-b328-38f3fb42dc2e

11.2.2 Bluetooth Name Compliance

CUP + XXX (Manufacture identifier) + X (Terminal Characteristic) + XXXXXXXX(The last 8 digits of Terminal Serial Number)

Terminal characteristic compliance is defined in below table:

Table 8 Terminal Characteristic Compliance

Terminal characteristic	Code
Dual mode + CCID	1
Dual mode + T1	2
Single mode 3.0 + CCID	3
Single mode 3.0 + T1	4
Single mode 4.0 + CCID	5
Single mode 4.0 + T1	6

11.2.3 Other compliance

Under RSA-1024 certificate system, the interaction time between bluetooth 4.0 and SecurePay gateway should be within 5 seconds

11.3 USB OTG Interface Protocols

Android USB OTG interface, Apple Lightning, Apple 30-pin dock interface, refer to 11.1 USB interface protocol to implement.

11.4 Other Interface Protocols

Other interface protocols are not specifically defined in these Specifications.

Appendix A

(Normative Appendix)

Terminal Command Set

A.1 Terminal Command Set Overview

Refer to UICS for the APDU format of terminal command and response..

The terminal command set is divided into two parts; special commands and normal commands.

- **Special Command Set:** refers to all APDU commands with message-type (CLA) 7E or 7F mentioned within these specifications. 7E refers to plaintext transmission data; 7F refers to encrypted transmission data, keys (including session keys (for data encryption) and MAC keys; keys are generated by mutual consensus using secure channels. Terminal special commands are divided into five categories based on their function: secure channel commands, transaction commands, issuer retention commands, and specification retention commands. For special command definitions, see the table below.
- **Normal command set:** refers to all other APDU commands aside from the special commands described in these specifications.

Notes:

- 1) **Management Commands:** Define the retrieval of terminal parameter data, control of terminal notifications, etc.
- 2) **Secure Channel Commands:** Defines secure channels created between terminals and processing centers, management of electronic certificates, etc.
- 3) **Transaction Commands:** Defines initiation and online processing commands of some debit/credit applications transactions
- 4) **Issuer Retention Commands:** Usage defined by issuer.
- 5) **Current Specification Retention Commands:** Reserved for future use.

Table A.1 Terminal Special Command Definitions

CLA Command Type	INS Instruction Code	Usage
7E/7F	10-1F	Management command
7E/7F	20-2F	Secure channel command
7E/7F	40-4F	Transaction command
7E/7F	50-6F	Issuer Retention

CLA Command Type	INS Instruction Code	Usage
7E/7F	Other	Current Specification Retention

For all special command sets mentioned in these specifications that are not specifically defined, datatype is defined by UICS. For terminal commands defined by these specifications, see the following table:

Table A.2 Terminal Special Command Set

No.	Command Name	CL A	IN S	Description of Function	Mandatory(M)/Optional(O)
Management Commands	READ TERMINAL INFO	7E	10	Used to retrieve terminal data, transaction IC card numbers, hardware version numbers, device status, etc.	M
	MANAGE BUZZER	7E	11	Used to control buzzer status	O
	MANAGE LED	7E	12	Used to control LED status	O
	CONFIG DISPLAY FORMAT	7E	13	Used to control display indicator information and format	O
	EXCHANGE STATUS	7E	14	Used for exchange of processing center and terminal statuses.	M
	GET CARD TERMINAL RECODE	7E	15	Retrieval of terminal response data.	M
	MULTIPLE INSTRUCTION	7E	16	Multiple instruction processing command.	M
	ADD CERTIFICATE	7E	20	Used to install certificate into terminals.	M
Channel Commands	UPDATE CERTIFICATE	7E	21	Used to update certificates stored within a terminal.	M
	DELETE CERTIFICATE	7E	22	Use to retrieve (delete) certificates stored within a terminal.	M
	READ CERTIFICATE	7E	23	Used to read certificates stored within a terminal.	M
	GET CERT RESPONSE	7E	24	Used to read data returned due to READ CERTIFICATE command.	M

No.	Command Name	CL A	IN S	Description of Function	Mandatory(M)/Optional(O)
	GET CLIENT HELLO	7E	25	Used to retrieve terminal algorithm identifier and terminal random number for establishment of secure channel.	M
	HASH SERVER CERTIFICATE	7E	26	Used to conduct hash calculation of trusted server certificate information.	M
	VERIFY SERVER CERTIFICATE	7E	27	Used to verify legality of trusted server certificate.	M
	CLIENT SIGN	7E	28	Uses terminal private key to sign input data.	M
	EXPORT MASTERKEY	7E	29	Exports shared main key in ciphertext form.	M
	HMAC	7E	2A	Used to generate terminal MAC and verify processing center HMAC/dispersion process keys.	M
	TRANSMIT ENCRYPTED COMMAND	7F	2B	Used to transmit post-encryption APDU commands.	M
	CLOSE SECURE CHANNEL	7E	2C	Used to close secure channel.	M
	READ NON-PBOC PIN	7E	2D	Used to read non-PBOC financial IC card PINs.	O
Transaction Commands	CREDIT FOR LOAD	7E	40	Used for online processing of financial IC card load transactions.	M
	DEBIT FOR PURCHASE	7E	41	Used for online processing of financial IC card spending transactions.	M
	GET ELECTRONIC CASH BALANCE	7E	42	Used to inspect financial IC card electronic cash balances.	O

No.	Command Name	CL A	IN S	Description of Function	Mandatory(M)/Optional(O)
	GET PRIMARY BALANCE	7E	43	Used to inspect financial IC card primary account balance.	M
	GET DOL VALUE	7E	45	Used to retrieve data object content in accordance with inputted data object list.	M
	GET REVERSE INFO	7E	46	Used to retrieve reversal information or script execution result information.	M
	READ CARDHOLDER INFO	7E	47	Used to retrieve cardholder information and other transaction process information.	M
	GET CARD TRADE RECODE	7E	48	Used to retrieve financial IC card card transaction logs.	M
	GREDIT CARD PAYMENT	7E	49	Used for online processing of financial IC card debt repayment transactions.	M
	VERITY OFFLINE PIN	7E	4A	Used to initiate terminal verification of offline card PIN.	M
	TRANSFER	7E	4B	Used for online processing of financial IC card transfer transactions.	

A.2 Management Command

A.2.1 READ TERMINAL INFO Command

A.2.1.1 Definition and Scope

READ TERMINAL INFO command is used to acquire the terminal manufacturer information, including firmware version number, terminal status, etc.

A.2.1.2 Command Message

See Table A.3 for the encoding of READ TERMINAL INFO command message:

Table A.3 READ TERMINAL INFO Command Message

Code	Value
CLA	7E
INS	10
P1	00
P2	00: Read status of terminal device /01: Read information of terminal device
Lc	Not Available
Data	Not Available
Le	Refer to the descriptions

A.2.1.3 Data Field of Command Message

Data field of command message does not exist.

A.2.1.4 Data Field of Response Message**P2=0x00:**

It indicates to acquire the status of terminal, return 1-byte status information, Le=0x01. The definition of status byte is shown as follow:

Table A.4 Terminal Status Byte Definitions

B7	B6	B5	B4	B3	B2	B1	B0	Remarks
							1	Terminal data already set
							0	Terminal data not yet set
						1		Root CA certificate already installed
						0		Root CA certificate not yet installed
					X			Reserved
				1				PIN encryption certificate already installed

B7	B6	B5	B4	B3	B2	B1	B0	Remarks
				0				PIN encryption certificate not yet installed
			1					Terminal certificate already installed
			0					Terminal certificate not yet installed
	0	0						Terminal status: power on
	1	0						Terminal status: transaction allowed
	X	X						Other values are Reserved
X								Reserved

P2=0x01:

It indicates to acquire the terminal data, firmware version number and the number of financial IC card.

Output data is in TLV form. Response data is defined in the table below:

Table A.5 READ TERMINAL INFO Response Message Data Fields

Explanation	Length (in bytes)	Remarks
Terminal data label	1	Tag value: 01
Terminal data length	1	
Terminal Data	23	
Terminal firmware version number data label	1	Tag value: 02
Terminal firmware version number data length	1	
Terminal firmware version number	1-16	
Public key version number data label	1	Tag value: 03

Explanation	Length (in bytes)	Remarks
Public key version number data length	1	
Public key version number	2	00-99
IC card card number data label	1	Tag value: 04
IC card card number data length	1	
IC card card number	14-19	
Reversal identifier bit data label	1	Tag value: 05
Reversal identifier bit data length	1	
Reversal identifier bit	1	0x30: No reversal information, 0x31: Reversal information
Terminal type number data label	1	Tag value: 06
Terminal type number data length	1	
Terminal type number	1-16	
Terminal issuer institution code label	1	Tag value: 07
Terminal issuer institution code length	1	
Terminal issuer institution code	8	The issuer institution code indicates the institution that issued a given terminal, which is used to differentiate the notion - acquirer. Code rules are the same as those for "affiliated institution codes".

Explanation	Length (in bytes)	Remarks
PIN certificate serial number code label	1	Tag value: 08 Note: To guarantee certificate security, response messages will only contain this data field after a secure channel has been established.
PIN certificate serial number code length	1	
PIN certificate serial number code	VAL	A PIN certificate's "certificate serial number" data field is used to uniquely identify that PIN certificate; the MiniPay processing system uses this serial number to retrieve detailed information about that certificate and judge whether or not the PIN certificate requires upgrading.
CA certificate serial number label	1	Tag value: 09 CA certificates are the father certificates of terminal and trusted server certificates
CA certificate serial number length	1	
CA certificate serial number content	VAL	'Serial Number' filed in CA certificate is unique among all CA certificate, MiniPay processing system uses CA certificate serial number and PIN certificate serial number to evaluate if upgrade is necessary
Initial transaction platform and communication interface code label	1	Label value '0x0A'
Initial transaction platform and communication interface code label	2	

Explanation	Length (in bytes)	Remarks
Initial transaction platform and communication interface code	2	Initial transaction platform and communication interface code is used to differentiate transaction platform and the used interface type, the first byte represents platform, the second byte represents communication interface, detailed format refer to Appendix K
Terminal certificate validation date label	1	Label value '0x0B', terminal certificate validation date
Terminal certificate validation date length	VAL	
Terminal certificate validation date content	VAL	asc code

A.2.1.5 Status Code of Response Message

The status code for successful execution of this command is '9000'

See Appendix A.5 List of Terminal Command Response Status Code for the error information which may be returned by the terminal

A.2.2 MANAGE BUZZER Command

A.2.2.1 Definition and Scope

MANAGE BUZZER command is used to control the state of buzzer. This command is an optional command.

A.2.2.2 Command Message

See Table A.6 for the MANAGE BUZZER command message encoding:

Table A.6 MANAGE BUZZER Command Message

Code	Value
CLA	7E
INS	11
P1	00
P2	00

Code	Value
Lc	01
Data	Buzzer status code
Le	Not Available

A.2.2.3 Data Field of Command Message

See Table A.7 for the data field encoding of command message:

Table A.7 MANAGE BUZZER Data Field Encoding

Value	Meanings
00	Close
01	Buzz one time
02	Continuously buzz
03	Intermittently buzz
Other	Reserved

A.2.2.4 Data Field of Response Message

The data field of response message doesn't exist.

A.2.2.5 Status Code of Response Message

The status code for successful execution of this command is '9000'.

See Appendix A.5 List of Terminal Command Response Status Code for the error information which may be returned by the terminal

A.2.3 MANAGE LED Command

A.2.3.1 Definition and Scope

The MANAGE LED command is used to set the status of LED. It reports the cardholder's transaction state in a simple and explicit way by setting the status of LED.

A.2.3.2 Command Message

See Table A.8 for the MANAGE LED command message encoding:

Table A.8 MANAGE LED Command Message

Code	Value
CLA	7E
INS	12
P1	00
P2	00
Lc	01
Data	LED status setting information
Le	Not Available

A.2.3.3 Data Field of Command Message

See Table A.9 for the data field encoding of command message:

Table A.9 MANAGE LED Data Field Encoding

Value	Meanings
00	Turn off LED
01	Light LED one time
02	Light LED for extended period
03	Blink LED
Other	Reserved

A.2.3.4 Data Field of Response Message

The data field of response message doesn't exist.

A.2.3.5 Status Code of Response Message

The status code for successful execution of this command is '9000'.

See Appendix A.5 List of Terminal Command Response Status Code for the error information which may be returned by the terminal

A.2.4 CONFIG DISPLAY FORMAT Command

A.2.4.1 Definition and Scope

CONFIG DISPLAY FORMAT is used to control display of terminal screen. The command is an optional command.

A.2.4.2 Command Message

See Table A.10 for the CONFIG DISPLAY FORMAT command message encoding:

Table A.10 CONFIG DISPLAY FORMAT Command Message

Code	Value
CLA	7E
INS	13
P1	Line number of displayed information
P2	Offset of displayed information
Lc	Length of displayed information
Data	Displayed information
Le	Not Available

Data field of command message

The data field of command message represents the character contents shown in the display.

A.2.4.3 Data Field of Response Message

The data field of response message doesn't exist.

A.2.4.4 Status Code of Response Message

The status code for successful execution of this command is '9000'

Refer to Appendix A.5 Terminal Command Response Status Code for the error information which may be returned by the terminal

A.2.5 EXCHANGE STATUS Command

A.2.5.1 Definition and Scope

EXCHANGE STATUS Command is used to exchange status between processing center and terminal

A.2.5.2 Command Messages

EXCHANGE STATUS command message codes are described in the table below:

Table A.11 EXCHANGE STATUS Command Messages

Code	Value
CLA	7E
INS	14
P1	Status (status exchange between processing center and terminal)
P2	Reserved
Lc	Not Available
Data	Not Available
Le	00

A.2.5.3 Command Message Data Fields

Does not exist.

A.2.5.4 Response Message Data Fields

Does not exist.

A.2.5.5 Response Message Status Codes

“90XX” designates that this command was executed successfully.

Refer to Appendix A.5 – Terminal Command Response Status Code Table for possible error codes for this command.

Note: For 0x90XX, XX is the transmission value of P1. For example: 0x7E 47 01 00 00; return: 0x9001

A.2.6 GET TERMINAL RESPONSE Command

A.2.6.1 Definition and Scope

When a transaction instruction returns 61XX, the server will use this instruction to retrieve terminal response data.

A.2.6.2 Command Messages

See the table below for information on GET CARD TERMINAL RESPONSE command message codes:

Table A.12 GET TERMINAL RESPONSE Command Messages

Code	Value
CLA	7E
INS	15
P1	00
P2	00
Lc	Not Available
Data	Not Available
Le	Expected response data length

A.2.6.3 Command Message Data Fields

Command message data fields do not exist.

A.2.6.4 Response Message Data Fields

Response message data field length is determined by the Le value.

A.2.6.5 Response Message Status Codes

“90XX” or “61XX” designates that this command was executed successfully.

Refer to Appendix A.5 – Terminal Command Response Status Code Table for possible error codes for this command.

A.2.7 MULTIPLE INSTRUCTION Command

A.2.7.1 Definition and Scope

The MULTIPLE INSTRUCTION processing command is used to bundle multiple card instructions; the terminal will analyze these instructions and execute them one by one until either all commands have been executed successfully or an error appears, then return either the execution result or error of the last executed instruction.

A.2.7.2 Command Messages

Refer to the table below for information on MULTIPLE INSTRUCTION command message codes:

Table A.13 MULTIPLE INSTRUCTION Command Messages

Code	Value
CLA	7E
INS	16
P1	00
P2	00
Lc	Data field byte length
Data	Multiple instruction contents
Le	'00'

A.2.7.3 Data Field of Command Message

Command message data fields include either several terminal instructions or several card instructions with each command separated by a “,”. During the instruction execution process, should execution of a command be unsuccessful then the terminal will halt the process and transmit the command serial number and response code of the last executed command.

For example: 7E160000 18 00A4040007A0000003330101 2C 0084000004 2C 0084000008

A.2.7.4 Response Message Data Fields

The response message data field is the response data of the last instruction of a multiple instruction command; should command execution fail, then the instruction serial code and card error response code will be returned.

A.2.7.5 Response Message Status Codes

“9000” designates that this command was executed successfully.

Refer to Appendix A.5 – Terminal Command Response Status Code Table for possible error codes for this command.

A.3 Secure Channel Command

A.3.1 ADD CERTIFICATE Command

A.3.1.1 Definition and Scope

ADD CERTIFICATE is used to add the public key certificate for the terminal.

A.3.1.2 Command Message

See Table A.14 for the ADD CERTIFICATE command message encoding:

Table A.14 ADD CERTIFICATE Command Message

Code	Value
CLA	7E
INS	20
P1	High 4bit represents certificate type and low 4bit represents offset value
P2	Offset value (low 8bit)
Lc	Certificate bundle length
Data	Certificate bundle data
Le	Not Available

The high 4-bit information of P1 defines the certificate type, is defined as follows:

Table A.15 Definition of Certificate Types

B7	B6	B5	B4	Explanation
----	----	----	----	-------------

B7	B6	B5	B4	Explanation
0	0	0	1	Level 1 Root CA Certificate
0	0	1	0	Level 2 Root CA Certificate
0	1	0	0	PIN Encryption Certificate
1	0	0	0	Reserved

The low 4-bit of P1 and the P2 byte form the 12-bit shift byte and the shift range is 0-4095 bytes.

A.3.1.3 Data Field of Command Message

Command message data field content includes certificate datagrams.

Limited by communication agreements, certificate data must be transmitted in segments; this means that ADD CERTIFICATE commands probably will be executed multiple times.

A.3.1.4 Data Field of Response Message

The data field of response message doesn't exist.

A.3.1.5 Status Code of Response Message

The status code for successful execution of this command is '9000'.

Refer to Appendix A.5 – Terminal Command Response Status Code Table for possible error codes for this command.

A.3.2 UPDATE CERTIFICATE Command

A.3.2.1 Definition and Scope

UPDATE CERTIFICATE command is used to update the existing public key certificate in the terminal

A.3.2.2 Command Message

Refer to Table A.16 for the UPDATE CERTIFICATE command message encoding:

Table A.16 UPDATE CERTIFICATE Command Message

Code	Value
CLA	7E

Code	Value
INS	21
P1	High 4bit represents certificate type and low 4bit represents offset value
P2	Offset value (low 8bit)
Lc	Certificate bundle length
Data	Certificate bundle data
Le	Not Available

The high 4-bit information of P1 defined enhanced certificate type is defined as follows:

Table A.17 Definition of Certificate Types

B7	B6	B5	B4	Explanation
0	0	0	1	Level 1 Root CA Certificate
0	0	1	0	Level 2 Root CA Certificate
0	1	0	0	PIN Encryption Certificate
1	0	0	0	Terminal Certificate

The low 4-bit of P1 and the P2 byte form the 12-bit shift byte and the shift range is 0-4095 bytes.

A.3.2.3 Data Field of Command Message

The content of command message data field includes the data of new certificate.

As limited by the maximum number of bytes in one communication, it is possible to transfer certificate data by splitting it into messages and execute Add CERTIFICATE for several times.

A.3.2.4 Data Field of Response Message

The data field of response message doesn't exist.

A.3.2.5 Status Code of Response Message

The status code for successful execution of this command is '9000'.

Refer to Appendix A.5 List of Terminal Command Response Status Code for the error information which may be returned by the terminal

A.3.3 DELETE CERTIFICATE Command

A.3.3.1 Definition and Scope

DELETE CERTIFICATE command is used to delete the existing public key certificate in the terminal

A.3.3.2 Command Message

Refer to Table A.18 for the DELETE CERTIFICATE command message encoding:

Table A.18 DELETE CERTIFICATE Command Message

Code	Value
CLA	7E
INS	22
P1	Certificate type identifier (Refer to explanation of Appendix A.17)
P2	00
Lc	Data field data length
Data	Terminal information
Le	Not Available

The high 4-bit information of P1 defined enhanced certificate type is defined as follows:

Table A.19 Definition of Certificate Types

B7	B6	B5	B4	Explanation
0	0	0	1	Level 1 Root CA Certificate
0	0	1	0	Level 2 Root CA Certificate

B7	B6	B5	B4	Explanation
0	1	0	0	PIN Encryption Certificate
1	0	0	0	Reserved

A.3.3.3 Command Message Data Fields

Terminals conduct verification of the information sent by the processing center; if verification results are the same, then the terminal will delete the selected certificate category; otherwise this command execution will fail.

Table A.20 DELETE CERTIFICATE Command Data Fields

Data	Length (in bytes)	Remarks
Terminal number	23	
Hardware version number	1-16	

A.3.3.4 Data Field of Response Message

The data field of response message doesn't exist.

A.3.3.5 Status Code of Response Message

The status code for successful execution of this command is '9000'.

Refer to Appendix A.5 List of Terminal Command Response Status Code for the error information which may be returned by the terminal

A.3.4 READ CERTIFICATE Command

A.3.4.1 Definition and Scope

READ CERTIFICATE is used to read the existing public key certificate in the terminal.

A.3.4.2 Command Message

Refer to Table A.21 for the READ CERTIFICATE command message encoding:

Table A.21 READ CERTIFICATE Command Message

Code	Value
CLA	7E

Code	Value
INS	23
P1	00: Read the certificate; 01: read certificate hash value
P2	See explanation of control parameters
Lc	Not Available
Data	Not Available
Le	See Description

Table A.22 Control Parameter Set Value Description

P2	Read content	Remarks
0x00	Terminal Certificate	
0x01	Level 1 Root CA Certificate	
0x02	Level 2 Root CA Certificate	
0x03	PIN Encryption Certificate	

A.3.4.3 Command Message Data Fields

Command message data field does not exist.

A.3.4.4 Response Message Data Fields

Response message data field does not exist.

A.3.4.5 Response Message Status Codes

A "61FF" code means that this command has been executed successful and indicates that GET CERT RESPONSE should be used to read response data.

Refer to Appendix A.5 – Terminal Command Response Status Code Table for possible error codes for this command.

A.3.5 GET CERT RESPONSE Command

A.3.5.1 Definition and Scope

The GET CERT RESPONSE Command is used to read the response data as returned by READ CERTIFICATE

A.3.5.2 Command Message

Refer to Table A.23 for the GET CERT RESPONSE command message encoding:

Table A.23 GET CERT RESPONSE Command Message

Code	Value
CLA	7E
INS	24
P1	00
P2	00
Lc	Not Available
Data	Not Available
Le	Expected certificate data bundle length

A.3.5.3 Data Field of Command Message

The data field of command message doesn't exist.

A.3.5.4 Data Field of Response Message

It is the certificate data returned as per the specified length.

A.3.5.5 Status Code of Response Message

"9000" indicates that this command has been executed successfully; a "61XX" code indicates that XX amount of data remains to be read.

Refer to Appendix A.5 – Terminal Command Response Status Code Table for possible error codes for this command.

A.3.6 GET CLIENT HELLO Command

A.3.6.1 Definition and Scope

GET CLIENT HELLO is used to acquire algorithm identifier and random number supported by the terminal

A.3.6.2 Command Message

Refer to Table A.24 for the GET CLIENT HELLO command message encoding:

Table A.24 GET CLIENT HELLO Command Message

Code	Value
CLA	7E
INS	25
P1	00
P2	00
Lc	Not exist
Data	Not exist
Le	0x21

A.3.6.3 Data Field of Command Message

The data field of command message does not exist.

A.3.6.4 Data Field of Response Message

Response data includes 0x01-byte algorithm identifiers and 0x20-byte random numbers. Refer to the table below for algorithm descriptions.

Table A.25 Byte Definition of Algorithm Descriptor

B7	B6	B5	B4	B3	B2	B1	B0	Algorithm
*	*	*	*	*	*	*	1	RSA
*	*	*	*	*	*	1	*	ECC
*	*	*	1	*	*	*	*	3DES
*	*	1	*	*	*	*	*	Reserved
*	1	*	*	*	*	*	*	1-CA 2048, 0-CA 1024
1	*	*	*	*	*	*	*	1-Reserved, 0-RSA

A.3.6.5 Status Code of Response Message

The status code for successful execution of this command is '9000'.

Refer to Appendix A.5 – Terminal Command Response Status Code Table for possible error codes for this command.

A.3.7 HASH SERVER CERTIFICATE Command

A.3.7.1 Definition and Scope

The HASH SERVER CERTIFICATE command is used to conduct summary operations on trusted server certificates; this summary result is compared against the decryption result of the VERIFY CERTIFICATE command.

A.3.7.2 Command Message

See Table A.26 for the HASH SERVER CERTIFICATE command message encoding:

Table A.26 HASH SERVER CERTIFICATE Command Message

Code	Value
CLA	7E
INS	26
P1	00
P2	00: SHA-1 algorithm; Other; retain
Lc	Length of input data
Data	Input data
Le	Not Available

A.3.7.3 Data Field of Command Message

The structure of input data is as follow: 1-byte flag bit + 1-byte extracting information offset value + server certificate message data

The 1-byte flag bit is defined as follow:

Table A.27 Meaning of Various Bits of the Mark Bit

B7	Flag bit for starting message of segmented certificate information
----	--

B6	Flag bit for ending message of segmented certificate information
B5	Flag bit for OU field to be extracted in the current data
B4	Flag bit for public key field to be extracted in the current data
B3	Reserved
B2	Reserved
B1	Reserved
B0	Reserved

Byte for 1-byte extracting information offset value: If it is required to extract OU field or public key field in the message data, it indicates the shift length from the first byte in the message data, so that the terminal firmware can rapidly extract this information.

A.3.7.4 Data Field of Response Message

The response message data does not exist.

A.3.7.5 Status Code of Response Message

The status code for successful execution of this command is '9000'.

Refer to Appendix A.5 – Terminal Command Response Status Code Table for possible error codes for this command.

A.3.8 VERIFY SERVER CERTIFICATE Command

A.3.8.1 Definition and Scope

The VERIFY SERVER CERTIFICATE command is used to decrypt trusted server certificate signature values; the decryption result is compared against the HASH SERVER CERTIFICATE summary value within the terminal; if the values are the same then the trusted server certificate is passed.

A.3.8.2 Command Message

Refer to Table A.28 for the VERIFY SERVER CERTIFICATE command message encoding:

Table A.28 VERIFY SERVER CERTIFICATE Command Message

Code	Value
CLA	7E
INS	27
P1	00
P2	00: RSA algorithm; Other; Reserved
Lc	Length of input data
Data	Input Data
Le	Not Available

A.3.8.3 Data Field of Command Message

The structure of input data is as follow: 1-byte flag bits + 1-byte offset value + signature value message data

The byte for 1-byte flag bits are defined as follow:

Table A.29 Meaning of Various Bits of the Mark Bit

B7	Flag bit for starting message of signature value
B6	Flag bit for ending message of signature value
B5	Reserved
B4	Reserved
B3	Reserved
B2	Reserved
B1	Reserved
B0	Reserved

As the transferred certificate signature value is in the TLV encoding format, the offset value represents the offset length of Value from the starting address of message

A.3.8.4 Data Field of Response Message

The response message data doesn't exist.

A.3.8.5 Status Code of Response Message

The status code for successful execution of this command is '9000'.

Refer to Appendix A.5 – Terminal Command Response Status Code Table for possible error codes for this command

A.3.9 CLIENT SIGN Command

A.3.9.1 Definition and Scope

The CLIENT SIGN Command uses a terminal private key to sign designated data and return a signature value.

A.3.9.2 Command Message

Refer to Table A.30 for the CLIENT SIGN command message encoding:

Table A.30 CLIENT SIGN Command Message

Code	Value
CLA	7E
INS	28
P1	00
P2	00
Lc	Length of input data
Data	Input data
Le	00

A.3.9.3 Data Field of Command Message

For source data please refer to Section 8.3.1.

A.3.9.4 Data Field of Response Message

Signed data

A.3.9.5 Status Code of Response Message

The status code for successful execution of this command is '9000'.

Refer to Appendix A.5 List of Terminal Command Response Status Code for the error information which may be returned by the terminal

A.3.10 EXPORT MASTERKEY Command

A.3.10.1 Definition and Scope

EXPORT MASTERKEY Command is used to acquire master key generated by the terminal, and use the public key of the trusted server certificate for encryption.

A.3.10.2 Command Message

See Table A.31 for the EXPORT MASTERKEY command message encoding:

Table A.31 EXPORT MASTERKEY Command Message

Code	Value
CLA	7E
INS	29
P1	00
P2	00
Lc	Not Available
Data	Not Available
Le	00

A.3.10.3 Data Field of Command Message

The command message data does not exist.

A.3.10.4 Data Field of Response Message

Master key information encrypted by the public key of trusted server certificate.

A.3.10.5 Status Code of Response Message

The status code for successful execution of this command is '9000'.

Refer to Appendix A.5 – Terminal Command Response Status Code Table for possible error codes for this command.

A.3.11 HMAC Command

A.3.11.1 Definition and Scope

HMAC command is used to:

- 1) Extract the HMAC value generated after the handshake process with terminal, which is then sent to the processing center for verification.
- 2) Input the HMAC value generated in the process of handshake with the processing center, which is then sent to the terminal for verification.
- 3) The terminal generates Session Key internally via HMAC algorithm.

Refer to the detailed definition of protocol process in Section 8.2.1 and 8.2.2 in the specification for the details of above contents.

A.3.11.2 Command Message

See Table A.32 for the HMAC command message encoding:

Table A.32 HMAC Command Message

Code	Value
CLA	7E
INS	2A
P1	00
P2	Control parameters (see explanation)
Lc	Set value according to P2, see explanation below
Data	Set value according to P2, see explanation below
Le	Set value according to P2, see explanation below

Table A.33 Control Parameter Set Value Table

P2	Lc	Data	Le	Remarks
0x00	Not Available	Not Available	0x14	HMAC value generated by completion of terminal handshake

0x01	0x14	Processing center HMAC value	0x00	HMAC value will be verified by terminal, which is generated after the processing center handshake completion
0x02	Not Available	Not Available	0x00	HMAC algorithm internal generation of session key

A.3.11.3 Data Field of Command Message

Refer to Table A.33 Control parameter set value table

A.3.11.4 Response Message Data Fields

When P2=00, the response message data field will be the HMAC value calculated by the terminal. Refer to Section 8.2.1 for details regarding HMAC data elements generated by terminals.

When P2=01 and P2=02, there are no response message data fields.

A.3.11.5 Status Code of Response Message

The status code for successful execution of this command is '9000'.

Refer to Appendix A.5 – Terminal Command Response Status Code Table for possible error codes for this command.

A.3.12 TRANSMIT ENCRYPTED COMMAND Command

A.3.12.1 Definition and Scope

The TRANSMIT ENCRYPTED COMMAND command is used for encrypted transmissions between the processing center and terminals.

A.3.12.2 Command Message

Refer to Table A.34 for the TRANSMIT ENCRYPTED COMMAND Command Message Encoding:

Table A.34 TRANSMIT ENCRYPTED COMMAND Command Message

Code	Value
CLA	7F
INS	2B
P1	00
P2	00: Non-cascade mode; 01: Cascade mode

Code	Value
Lc	Length of input data
Data	Input data
Le	00

For P2 = 0x00, the information message of secure channel is transmitted in the non-cascade mode or the information message has reached the last frame of cascade data

For P2 = 0x01, the information message of secure channel is transmitted in the non-cascade mode and the succeeding data is present

A.3.12.3 Data Field of Command Message

SKey encrypted command and MAC Key calculated with MKey from processing center.

A.3.12.4 Data Field of Response Message

Encrypted command response data and MAC

A.3.12.5 Status Code of Response Message

The status code for successful execution of this command is '9000'.

Refer to Appendix A.5 List of Terminal Command Response Status Code for the error information which may be returned by the terminal.

Notes:

- (1) Transaction commands can only be transmitted to terminals using this encryption command; that is to say that terminal transactions can only be conducted after a secure channel has been established.
- (2) Terminal commands are transmitted using this command, and will be executed after decryption; execution results and response codes must be encrypted before they are sent to the processing center, and return codes must be sent directly after return ciphertexts in plaintext format.

Example: the '【】' (brackets) below indicate APDU command data fields; this field includes two parts; the encrypted plaintext data within the brackets as well as MAC ciphertext. Refer to Section 8.2.2 for a specific explanation of how the content of this data field is calculated.

Processing center command: 7F 2B 00 00 10 【00 05 7E 2C 00 00 00】

Terminal return data: 【90 00】 90 00 90 00

A.3.13 CLOSE SECURE CHANNEL Command

A.3.13.1 Definition and Scope

CLOSE SECURE CHANNEL commands are used to close secure channels and destroy all keys used within that channel.

A.3.13.2 Command Messages

See the table below for information on CLOSE SECURE CHANNEL command message codes:

Table A.35 CLOSE SECURE CHANNEL Command Messages

Code	Value
CLA	7E
INS	2C
P1	00
P2	00
Lc	Not Available
Data	Not Available
Le	00

A.3.13.3 Data Field of Command Message

The data field of command message does not exist.

A.3.13.4 Status Code of Response Message

The data field of response message does not exist.

A.3.13.5 Status Code of Response Message

The status code for successful execution of this command is '9000'.

Refer to Appendix A.5 – Terminal Command Response Status Code Table for possible error codes for this command.

A.3.14 READ NON-PBOC PIN Command

A.3.14.1 Definition and Scope

READ NON-PBOC PIN is used to read the PIN of non-UICS financial IC card, such as industry card PIN. The PIN of financial IC card or financial account is not allowed to be read. This command is an optional command.

A.3.14.2 Command Message

See Table A.36 for READ NON-PBOC PIN command message coding:

Table A.36 READ NON-PBOC PIN Command Message

Code	Value
CLA	7E/7F
INS	2D
P1	00
P2	00
Lc	Not Available
Data	Not Available
Le	Expected PIN length

A.3.14.3 Data Field of Command Message

The data field of command message does not exist.

A.3.14.4 Data Field of Response Message

The data field of response message contains six-digit of non-financial IC card PIN.

A.3.14.5 Status Code of Response Message

The status code for successful execution of this command is '9000'.

Refer to Appendix A.5 – Terminal Command Response Status Code Table for possible error codes for this command.

A.4 Transaction Command

A.4.1 CREDIT FOR LOAD Command

A.4.1.1 Definition and Scope

The CREDIT FOR LOAD command is used to support financial IC card online load transactions; permits financial IC card primary account funds to be transferred to an electronic cash account and completes IC card electronic cash balance update operations. Load amounts are inputted at the terminal. During the transaction the terminal will automatically check load limits and inform the cardholder of the maximum allotted load amount.

A.4.1.2 Command Message

For CREDIT FOR LOAD command message encoding please refer to Table A.37:

Table A.37 CREDIT FOR LOAD Command Message

Code	Value
CLA	7E
INS	40
P1	00/01
P2	When P1=00 and P2=00: Start transaction, read data as defined in Table A.36; When P1=00 and P2=01: read data as defined in Table A.37; When P1=01 and P2=00: Online response data
Lc	Length of Data field
Data	Refer to explanation of command message data fields
Le	00

A.4.1.3 Command Message Data Fields

When P1=00 and P2=00:

Start load transaction; see the command message data field table below:

Table A.38 Start Load Transaction Command Message Data Fields

Data	Length (in bytes)	Remarks
Transaction Amount	6	Set to 0
Transaction Date YYMMDD	3	Current date at processing center
Transaction Time HHMMSS	3	Current time at processing center
Other Transaction Data	Variable (VAR)	See notes below table
“Other transaction data” finish indicator (Tag FF01)	11	Content set as “\x12\x34\x56\x78\x90\xAB\xCD\xEF”
Minimum load amount (Tag FF02)	9	TLV

Notes: Other transaction date refers to processing center data; during abnormal transactions (such as reversals, script notifications, etc.) this data must be sent back to the processing center in abnormal (such as reversals and script notifications) transaction messages. Terminals do not need to analyze this data. Other transaction processing methods are the same.

When P1=00 and P2=01:

The command message does not contain data fields.

When P1=01:

Online returned data; see the table below for command message data fields:

Table A.39 Load Online Data Command Message Data Fields

Data	Length (in bytes)	Remarks
Online Connection Result	1	00: Normal connection; 01: Unable to connect
Issuer Authorization Data (Tag 91)	10-18	TLV format
Authorization Response Code (Tag 8A)	4	TLV format
71 Script Data (Tag 71)	Variable	TLV format

Data	Length (in bytes)	Remarks
72 Script Data (Tag 72)	Variable	TLV format

A.4.1.4 Response Message Data Fields

When P1=00 and P2=00:

Start load transaction; see the response message data field table below:

Table A.40 Start Load Transaction Response Message Data Fields

Data	Length (in bytes)	Remarks
Load amount	6	Actual load amount
Transaction result	1	01: Transaction declined; 02: Request for online transaction
Terminal Verification Result (TVR) (Tag 95)	7	TLV format
Transaction Date (Tag 9A)	5	TLV format
Random Number (Tag 9F37)	7	TLV format
Application Interchange Profile (AIP) (Tag 82)	4	TLV format
Application Transaction Code (ATC) Tag 9F36)	5	TLV format
Ciphertext Information Type (CID) (Tag 9F27)	4	TLV format
Application Ciphertext (AC) (Tag 9F26)	11	TLV format
Issuer Application Data (Tag 9F10)	Maximum 35	TLV format
Cardholder Verification Method (CVM) Result (Tag 9F34)	6	TLV format

Data	Length (in bytes)	Remarks
Transaction Sequence Counter (Tag 9F41)	5-7	TLV format
Specialized Document Name (Tag 84)	7-18	TLV format
Application Version Number (Tag 9F09)	5	TLV format
Authorized Balance (Tag 9F02)	9	Actual load balance

When P1=00 and P2=01:

Read PAN, PAN serial number, and related information: see the command response data field table below:

Table A.41 Response Message Data Fields

Data	Length (in bytes)	Remarks
PAN (Tag 5A)	Maximum 12	Application primary account, TLV format
PAN Serial Number (Tag 5F34)	4	Application primary account serial number, TLV format
Magstrip-Equivalent Data (Tag 57)	Maximum 21	Magstripe track 2 equivalent data, TLV format
Online PIN Ciphertext (Tag 99)	130	Ciphertext data encrypted by PIN

When P1=01:

Returned online data sent from terminal to processing center; see the response message data field table below:

Table A.42 Load Online Response Message Data Fields

Data	Length (in bytes)	Remarks
Transaction result	1	00: Transaction approved; 01: Transaction declined
Script Execution Result (Tag DF31)	8	TLV format
Terminal Verification Result (TVR) (Tag 95)	7	TLV format
Application Transaction Code (ATC) (Tag 9F36)	5	TLV format
Ciphertext Information Type (CID) (Tag 9F27)	4	TLV format
Application Ciphertext (AC) (Tag 9F26)	11	TLV format
Issuer Application Data (Tag 9F10)	Maximum 35	TLV format

A.4.1.5 Status Code of Response Message

The status code for successful execution of this command is '9000'.

Refer to Appendix A.5 – Terminal Command Response Status Code Table for possible error codes for this command.

A.4.2 DEBIT FOR PURCHASE Command

A.4.2.1 Definition and Scope

DEBIT FOR PURCHASE command is used to support online purchase transactions of financial IC card, allowing the cardholder to use financial IC card to complete purchases on the Internet and receive related services.

A.4.2.2 Command Message

For DEBIT FOR PURCHASE command message encoding please refer to Table A.43:

Table A.43 DEBIT FOR PURCHASE Command Message

Code	Value
CLA	7E
INS	41
P1	00/01
P2	When P1=00 and P2=00: Start transaction, read data as defined in Table A.42; When P1=00 and P2=01: read data as defined in Table A.43; When P1=01 and P2=00: Online data return
Lc	Data field data length
Data	See explanation of command message data fields
Le	00

A.4.2.3 Command Message Data Fields**When P1=00 and P2=00:**

Start online purchase transaction; see the command message data field table below:

Table A.44 Command Message Data Fields

Data	Length (in bytes)	Remarks
Transaction Amount	6	Spending amount
Transaction Date YYMMDD	3	Current date at processing center
Transaction Time HHMMSS	3	Current time at processing center
Other Transaction Data	Variable (VAR)	See notes below table

Notes: Other transaction date refers to processing center data; during abnormal transactions (such as reversals, script notifications, etc.) this data must be sent back to the processing center in abnormal (such as reversals and script notifications) transaction messages. Terminals do not need to analyze this data.

When P1=00 and P2=01:

The command message does not contain data fields.

When P1=01:

Online returned data; see the table below for command message data fields:

Table A.45 Command Message Data Fields

Data	Length (in bytes)	Remarks
Online Connection Result	1	00: Normal connection; 01: Unable to connect
Issuer Authorization Data (Tag 9F)	10-18	TLV format
Authorization Response Code (Tag 8A)	4	TLV format
71 Script Data (Tag 71)	Variable	TLV format
72 Script Data (Tag 72)	Variable	TLV format

A.4.2.4 Response Message Data Fields**When P1=00 and P2=00:**

Start online purchase transaction; see the command message data field table below:

Table A.46 Response Message Data Fields

Data	Length (in bytes)	Remarks
Transaction result	1	01: Transaction declined; 02: Request Online connection
Terminal Verification Result (TVR) (Tag 95)	7	TLV format
Transaction Date (Tag 9A)	5	TLV format
Random Number (Tag 9F37)	7	TLV format
Application Interchange Profile (AIP) (Tag 82)	4	TLV format

Data	Length (in bytes)	Remarks
Application Transaction Code (ATC) Tag 9F36)	5	TLV format
Ciphertext Information Type (CID) (Tag 9F27)	4	TLV format
Application Ciphertext (AC) (Tag 9F26)	11	TLV format
Issuer Application Data (Tag 9F10)	Maximum 35	TLV format
Cardholder Verification Method (CVM) Result (Tag 9F34)	6	TLV format
Transaction Sequence Counter (Tag 9F41)	5-7	TLV format
Special File Name (Tag 84)	7-18	TLV format
Application Version Number (Tag 9F09)	5	TLV format

When P1=00 and P2=01:

Read PAN, PAN serial number, and related information: see the command response data field table below:

Table A.47 Response Message Data Fields

Data	Length (in bytes)	Remarks
PAN (Tag 5A)	Maximum 12	Application primary account number, TLV format
PAN Serial Number (Tag 5F34)	4	Application primary account number serial number, TLV format
Magstrip-Equivalent Data (Tag 57)	Maximum 21	Magstripe track 2 equivalent data, TLV format

Data	Length (in bytes)	Remarks
Online PIN Ciphertext (Tag 99)	130	online PIN ciphertext encrypted by PIN key

When P1=01:

Returned online data sent from terminal to processing center; see the response message data field table below:

Table A.48 Response Message Data Fields

Data	Length (in bytes)	Remarks
Transaction result	1	00: Transaction approved; 01: Transaction refused
Script Execution Result (Tag DF31)	8	TLV format
Terminal Verification Result (TVR) (Tag 95)	7	TLV format
Application Transaction Code (ATC) (Tag 9F36)	5	TLV format
Ciphertext Information Type (CID) (Tag 9F27)	4	TLV format
Application Ciphertext (AC) (Tag 9F26)	11	TLV format
Issuer Application Data (Tag 9F10)	Maximum 35	TLV format

A.4.2.5 Response Message Status Codes

“9000” designates that this command was executed successfully.

Refer to Appendix A.5 – Terminal Command Response Status Code Table for possible error codes for this command.

A.4.3 GET ELECTRONIC CASH BALANCE Command

A.4.3.1 Definition and Scope

GET ELECTRONIC CASH BALANCE command is used to inquire electronic cash balance of IC card via the terminal, and the transaction is operated offline.

A.4.3.2 Command Message

For GET ELECTRONIC CASH BALANCE command message encoding please refer to Table A.49:

Table A.49 GET ELECTRONIC CASH BALANCE Command Message

Code	Value
CLA	7E
INS	42
P1	00: Display electronic cash balance
P2	00
Lc	Not Exist
Data	Not Exist
Le	00

A.4.3.3 Data Field of Command Message

The data field of command message does not exist.

A.4.3.4 Response Message Data Fields

See table below for financial IC card electronic cash balance/response message data fields:

Table A.50 GET ELECTRONIC CASH BALANCE Response Message Data Fields

Explanation	Length (in bytes)	Remarks
Electronic Cash Balance	6	BCD Code

A.4.3.5 Status Code of Response Message

The status code for successful execution of this command is '9000'.

Refer to Appendix A.5 List of Terminal Command Response Status Code for the error information which may be returned by the terminal

A.4.4 GET PRIMARY BALANCE Command

A.4.4.1 Definition and Scope

GET PRIMARY BALANCE command is used to inquire primary debit/credit account balance via online channel.

A.4.4.2 Command Message

GET PRIMARY BALANCE Command Message encoding is shown in Table A.51:

Table A.51 GET PRIMARY BALANCE Command Message

Code	Value
CLA	7E
INS	43
P1	00/01
P2	When P1=00 and P2=00: Start transaction, read data as defined in Table A.50; When P1=00 and P2=01: read data as defined in Table A.51; When P1=01 and P2=00: Online data return
Lc	Data field data length
Data	See the description of the data field of command message
Le	00

A.4.4.3 Command Message Data Fields

When P1=00 and P2=00:

Begins debit/credit primary account balance inspection transaction; see command message data field table below:

Table A.52 Command Message Data Fields

Data	Length (in bytes)	Remarks
Transaction Amount	6	Balance set as 0

Data	Length (in bytes)	Remarks
Transaction Date YYMMDD	3	Current date at processing center
Transaction Time HHMMSS	3	Current time at processing center
Other Transaction Data	Variable (VAR)	See notes below table

Notes: Other transaction data is processing center data; during abnormal transactions (such as reversals, script notifications, etc.) this data must be sent back to the processing center in abnormal transaction messages (such as reversals and script notifications). Terminals do not need to analyze this data.

When P1=00 and P2=01:

The command message does not contain data fields.

When P1=01:

Online returned data; see the table below for command message data fields:

Table A.53 Command Message Data Fields

Data	Length (in bytes)	Remarks
Connection Result	1	00: Normal connection; 01: Unable to connect
Primary account Balance	6	BCD Debit card: Primary account balance Credit card: Used amount
Issuer Authorization Data (Tag 9F)	10-18	TLV format
Authorization Response Code (Tag 8A)	4	TLV format
71 Script Data (Tag 71)	Variable	TLV format
72 Script Data (Tag 72)	Variable	TLV format
Card Type (Tag FF01)	4	00: Debit card; 01: Credit card

Data	Length (in bytes)	Remarks
Available balance (Tag FF02)	9	TLV format Debit card: Does not have this data field Credit card: Available limit

Note: For debit cards, micropayment will only display "Primary Account Balance"; if the card is a debit card then the micropayment will only display "Available Balance", "Used Amount".

A.4.4.4 Response Message Data Fields

When P1=00 and P2=00:

Start debit/credit primary account balance inspection transaction; see command message data field table below:

Table A.54 Response Message Data Fields

Data	Length (in bytes)	Remarks
Transaction result	1	01: Transaction declined; 02: Online Connection request
Terminal Verification Result (TVR) (Tag 95)	7	TLV format
Transaction Date (Tag 9A)	5	TLV format
Random Number (Tag 9F37)	7	TLV format
Application Interchange Profile (AIP) (Tag 82)	4	TLV format
Application Transaction Code (ATC) Tag 9F36)	5	TLV format
Ciphertext Information Type (CID) (Tag 9F27)	4	TLV format
Application Ciphertext (AC) (Tag 9F26)	11	TLV format

Data	Length (in bytes)	Remarks
Issuer Application Data (Tag 9F10)	Maximum 35	TLV format
Cardholder Verification Method (CVM) Result (Tag 9F34)	6	TLV format
Transaction Sequence Counter (Tag 9F41)	5-7	TLV format
Special File Name (Tag 84)	7-18	TLV format
Application Version Number (Tag 9F09)	5	TLV format

When P1=00 and P2=01:

Read PAN, PAN serial number, and related information: see the command response data field table below:

Table A.55 Response Message Data Fields

Data	Length (in bytes)	Remarks
PAN (Tag 5A)	Maximum 12	Application primary account number
PAN Serial Number (Tag 5F34)	4	Application primary account serial number
Magstrip-Equivalent Data (Tag 57)	Maximum 21	Magstrip track 2 equivalent data
Online PIN Ciphertext (Tag 99)	130	online PIN ciphertext encrypted by PIN key

When P1=01:

Returned online data sent from terminal to processing center; see the response message data field table below:

Table A.56 Response Message Data Fields

Data	Length (in bytes)	Remarks
Transaction result	1	00: Transaction approved; 01: Transaction refused
Script Execution Result (Tag DF31)	8	TLV format
Terminal Verification Result (TVR) (Tag 95)	7	TLV format
Application Transaction Code (ATC) (Tag 9F36)	5	TLV format
Ciphertext Information Type (CID) (Tag 9F27)	4	TLV format
Application Ciphertext (AC) (Tag 9F26)	11	TLV format
Issuer Application Data (Tag 9F10)	Maximum 35	TLV format

A.4.4.5 Response Message Status Codes

The status code for successful execution of this command is '9000'.

Refer to Appendix A.5 – Terminal Command Response Status Code Table for possible error codes for this command.

A.4.5 GET DOL VALUE Command**A.4.5.1 Definition and Scope**

The GET DOL TAGE VALUE is used to retrieve data object content in accordance with inputted data object list.

A.4.5.2 Command Messages

See the table below for information on GET DOL VALUE command message codes:

Table A.57 GET DOL VALUE Command Messages

Code	Value
CLA	7E
INS	45
P1	00
P2	00
Lc	Data field length
Data	Refer to A.4.5.3 explanation of command message data fields
Le	00

A.4.5.3 Command Message Data Fields

See table below for input data object list format:

Table A.58 GET DOL VALUE Command Message Data Field Format

Tag 1	Length 1	Tag 2	Length 2		Tag n	Length n
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Note: Length indicates required length of read label value; see explanation of DOL format.

A.4.5.4 Response Message Data Fields

Response message data field are the data objects of BER-TLV code. These data objects must be in accordance with the following format codes:

Table A.59 GET DOL VALUE Response Message Data Field Format

Tag 1	Length 1	Value 1	Tag 2	Length 2	Value 2		Tag 3	Length 3	Value 3
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A.4.5.5 Response Message Status Codes

“9000” designates that this command was executed successfully.

Refer to Appendix A.5 – Terminal Command Response Status Code Table for possible error codes for this command.

A.4.6 GET REVERSAL INFO Command

A.4.6.1 Definition and Scope

The GET REVERSAL INFO command is used to retrieve financial IC card reversal information or script execution result notifications saved during abnormal online transactions.

A.4.6.2 Command Message

GET REVERSAL INFO Command Message encoding is shown in Table A.60:

Table A.60 GET REVERSAL INFO Command Message

Code	Value
CLA	7E
INS	46
P1	00: Read information; 01: Clear information
P2	00: Reversal information; 01: Script notification
Lc	Not Available
Data	Not Available
Le	00

A.4.6.3 Data Field of Command Message

The data field of command message does not exist.

A.4.6.4 Response Message Data Fields

When P1=00 and P2=00:

Read transaction reversal information; see table below for command response data fields:

Table A.61 Read Reversal Information Command Response Data Field Formats

Data	Length (in bytes)	Remarks
Transaction result	1	00: Transaction approved; 01: Transaction declined; 02: Request connection

Data	Length (in bytes)	Remarks
Transaction Amount	6	Original transaction amount
Transaction Date YYMMDD	3	Original transaction date
Transaction Time HHMMSS	3	Original transaction time
Other Transaction Data	Variable (VAR)	Data saved at time of transaction
PAN (Tag 5A)	Maximum 12	Application primary account number, TLV format
PAN Serial Number (Tag 5F34)	4	Application primary account serial number, TLV format
Magstrip-Equivalent Data (Tag 57)	Maximum 21	Magstrip track 2 equivalent data, TLV format
Script Execution Result (Tag DF31)	8	TLV format
Terminal Verification Result (TVR) (Tag 95)	7	TLV format
Application Transaction Code (ATC) (Tag 9F36)	5	TLV format
Ciphertext Information Type (CID) (Tag 9F27)	4	TLV format
Application Ciphertext (AC) (Tag 9F26)	11	TLV format
Issuer Application Data (Tag 9F10)	Maximum 35	TLV format

When P1=00 and P2=01:

Command message data fields do not exist.

When P1=01 and P2=00:

Read script execution result notification information; see table below command response data fields:

Table A.62 Read Script Execution Notification Command Response Data Field Formats

Data	Length (in bytes)	Remarks
Transaction result	1	00: Transaction approved; 01: Transaction refused; 02: Request connection
Transaction Amount	6	Original Transaction Amount
Transaction Date YYMMDD	3	Original Transaction Date
Transaction Time HHMMSS	3	Original transaction time
Other Transaction Data	Variable (VAR)	Data saved at time of transaction
PAN (Tag 5A)	Maximum 12	Application primary account number, TLV format
PAN Serial Number (Tag 5F34)	4	Application primary account serial number, TLV format
Magstrip-Equivalent Data (Tag 57)	Maximum 21	Magstripe track 2 equivalent data, TLV format
Script Execution Result (Tag DF31)	8	TLV format
Terminal Verification Result (TVR) (Tag 95)	7	TLV format
Application Transaction Code (ATC) (Tag 9F36)	5	TLV format
Ciphertext Information Type (CID) (Tag 9F27)	4	TLV format
Application Ciphertext (AC) (Tag 9F26)	11	TLV format
Issuer Application Data (Tag 9F10)	Maximum 35	TLV format

When P1=01 and P2=01:

Command message data fields do not exist.

A.4.6.5 Status Code of Response Message

The status code for successful execution of this command is '9000'.

Refer to Appendix A.5 – Terminal Command Response Status Code Table for possible error codes for this command.

A.4.7 READ CARDHOLDER INFO Command**A.4.7.1 Definition and Scope**

The READ CARDHOLDER INFO command is used to read from the device other information about the cardholder participating in the current transaction. The cardholder will input appropriate cardholder information in accordance with the instructions displayed by on the device display screen. Cardholder information includes cardholder phone number, cardholder ID number, cellphone dynamic password, etc; the cardholder must input and confirm this information during the transaction process.

A.4.7.2 Command Messages

See the table below for REVERSAL INFO command message codes:

Table A.63 READ CARDHOLDER INFO Command Messages

Code	Value
CLA	7E
INS	47
P1	Display information display parameters (See Table A.64)
P2	00 (Reserved)
Lc	Display length of user input data
Data	Display information
Le	00

Table A.64 Display Format Parameter Table

Display Format Code	Explanation
00	Align to upper-left, normal mode display
05	Align to upper-left, display “*”
10	Align to lower-left, normal mode display
15	Align to lower-left, display “*”
20	Align centrally to left, normal mode display
25	Align centrally to left, display “*”
40	Align to upper-right, normal mode display
45	Align to upper-right, display “*”
50	Align to lower-right, normal mode display
55	Align to lower-right, display “*”
60	Align centrally to right, normal mode display
65	Align centrally to right, display “*”
80	Align to upper center, normal mode display
85	Align to upper center, display “*”
90	Align to lower center, normal mode display
95	Align to lower center, display “*”
A0	Align to center, normal mode display
A5	Align to center, display “*”

A.4.7.3 Command Message Data Fields

Command message data fields do not exist.

A.4.7.4 Response Message Data Fields

X(1): Cardholder information length

X(n): Cardholder information (ASCII)

A.4.7.5 Response Message Status Codes

“9000” designates that this command was executed successfully.

Refer to Appendix A.5 – Terminal Command Response Status Code Table for possible error codes for this command.

A.4.8 GET CARD TRADE RECODE Command

A.4.8.1 Definition and Scope

The GET CARD TRADE RECODE command is used on IC card transaction detail record files.

A.4.8.2 Command Messages

See the table below for information on GET CARD TRADE RECODE command message codes:

Table A.65 GET CARD TRADE RECODE Command Messages

Code	Value
CLA	7E/7F
INS	48
P1	00
P2	Control Parameters (See response message data fields)
Lc	Not Exist
Data	Not Exist
Le	00

Le = '00': Indicates maximum byte number required (256 bytes)

A.4.8.3 Command Message Data Fields

Command message data fields do not exist.

A.4.8.4 Response Message Data Fields

When P2=01, response data is the card log record format data element.

When P2=00, response data format is as follows:

Data Field	Explanation
Total record length	2 bytes (length does not include number of records)
Number of Record N	1 Byte
Record 1 Record N	See UnionPay IC Card Specifications Part 5 – UICS for record file content format

Note: When response data is greater than 255 bytes then it must be transmitted in bundles and the response message status code will be 61XX (successive data length). When a server receives a status response of 61XX it must use the GET TERMINAL RESPONSE instruction to retrieve the record data and combine it into a complete message.

A.4.8.5 Response Message Status Codes

“90XX” or “61XX” designates that this command was executed successfully.

Refer to Appendix A.5 – Terminal Command Response Status Code Table for possible error codes for this command.

A.4.9 CREDIT CARD PAYMENT Command

A.4.9.1 Definition and Scope

Credit card payment transaction command.

A.4.9.2 Command Messages

See the table below for CREDIT CARD PAYMENT command message codes:

Table A.66 CREDIT CARD PAYMENT Command Messages

Code	Value
CLA	7E/7F
INS	49
P1	00/01

Code	Value
P2	When P1=00 and P2=00: Start transaction, read data as defined in Table A.69; When P1=00 and P2=01: read data as defined in Table A.70; When P1=01 and P2=00: Online data return
Lc	Data field data length
Data	See explanation of command message data fields
Le	'00

A.4.9.3 Command Message Data Fields

When P1=00 and P2=00:

Start credit payment transaction; see the command message data field table below:

Table A.67 Command Message Data Fields

Data	Length (in bytes)	Remarks
Transaction Amount	6	payment amount
Transaction Date YYMMDD	3	Current date at processing center
Transaction Time HHMMSS	3	Current time at processing center
Credit Card Account Number	28	Pad from right with spaces if not 28
Other Transaction Data	Variable (VAR)	See notes below table

Notes: Other transaction date refers to processing center data; during abnormal transactions (such as reversals, script notifications, etc.) this data must be sent back to the processing center in abnormal transaction messages (such as reversals and script notifications). Terminals do not need to analyze this data.

When P1=00 and P2=01:

The command message does not contain data fields.

When P1=01:

Online returned data; see the table below for command message data fields:

Table A.68 Command Message Data Fields

Data	Length (in bytes)	Remarks
Connection Result	1	00: Normal connection; 01: Unable to connect
Issuer Authorization Data (Tag 9F)	10-18	TLV format
Authorization Response Code (Tag 8A)	4	TLV format
71 Script Data (Tag 71)	Variable	TLV format
72 Script Data (Tag 72)	Variable	TLV format

A.4.9.4 Response Message Data Fields**When P1=00 and P2=00:**

Start credit card repayment transaction; see the command message data field table below:

Table A.69 Response Message Data Fields

Data	Length (in bytes)	Remarks
Transaction result	1	01: Transaction declined; 02: Online Connection request
Terminal Verification Result (TVR) (Tag 95)	7	TLV format
Transaction Date (Tag 9A)	5	TLV format
Random Number (Tag 9F37)	7	TLV format
Application Interchange Profile (AIP) (Tag 82)	4	TLV format
Application Transaction Code (ATC) Tag 9F36)	5	TLV format
Ciphertext Information Type (CID) (Tag 9F27)	4	TLV format

Data	Length (in bytes)	Remarks
Application Ciphertext (AC) (Tag 9F26)	11	TLV format
Issuer Application Data (Tag 9F10)	Maximum 35	TLV format
Cardholder Verification Method (CVM) Result (Tag 9F34)	6	TLV format
Transaction Sequence Counter (Tag 9F41)	5-7	TLV format
Special File Name (Tag 84)	7-18	TLV format
Application Version Number (Tag 9F09)	5	TLV format

When P1=00 and P2=01:

Read PAN, PAN serial number, and related information: see the command response data field table below:

Table A.70 Response Message Data Fields

Data	Length (in bytes)	Remarks
PAN (Tag 5A)	Maximum 12	Application primary account number, TLV format
PAN Serial Number (Tag 5F34)	4	Application primary account number serial number, TLV format
Magstrip-Equivalent Data (Tag 57)	Maximum 21	Magstrip track 2 equivalent data, TLV format
Online PIN Ciphertext (Tag 99)	130	PIN/public key encrypted-online PIN ciphertext

When P1=01:

Returned online data sent from terminal to processing center; see the response message data field table below:

Table A.71 Response Message Data Fields

Data	Length (in bytes)	Remarks
Transaction result	1	00: Transaction approved; 01: Transaction declined
Script Execution Result (Tag DF31)	8	TLV format
Terminal Verification Result (TVR) (Tag 95)	7	TLV format
Application Transaction Code (ATC) (Tag 9F36)	5	TLV format
Ciphertext Information Type (CID) (Tag 9F27)	4	TLV format
Application Ciphertext (AC) (Tag 9F26)	11	TLV format
Issuer Application Data (Tag 9F10)	Maximum 35	TLV format

A.4.9.5 Response Message Status Codes

“9000” designates that this command was executed successfully.

Refer to Appendix A.5 – Terminal Command Response Status Code Table for possible error codes for this command.

A.4.10 VERITY OFFLINE PIN Command**A.4.10.1 Definition and Scope**

The VERITY OFFLINE PIN command is used by the terminal to verify an offline PIN. The verification command causes the IC card to compare the transaction PIN data contained within the command message data field and the corresponding parameter PIN data. The verification method is determined by the IC card application.

A.4.10.2 Command Messages

See the table below for information on VERITY OFFLINE PIN command message codes:

Table A.72 VERITY OFFLINE PIN Command Messages

Code	Value
CLA	7E
INS	4A
P1	00
P2	00
Lc	Not Exist
Data	Not Exist
Le	00

A.4.10.3 Command Message Data Fields

Command message data fields do not exist.

A.4.10.4 Response Message Data Fields

Response message data fields do not exist.

A.4.10.5 Response Message Status Codes

“9000” designates that this command was executed successfully.

Refer to Appendix A.5 – Terminal Command Response Status Code Table for possible error codes for this command.

A.4.11 TRANSFER Command**A.4.11.1 Definition and Scope**

The TRANSFER command is used for account transfer transactions. Transfer transactions come in two modes: IC card transfers and card-not-present channel transfers.

A.4.11.2 Command Messages

TRANSFER command message codes are described in the table below:

Table A.73 TRANSFER Command Messages

Code	Value
CLA	7E
INS	4B
P1	00/01/02/03
P2	IC Card transfer procedures: When P1=00 and P2=00: Start transaction, read data as defined in Table A.79; When P1=00 and P2=01: read data as defined in Table A.80; When P1=01 and P2=00: Online data return Card-not-present channel transfer procedures: When P1=02 and P2=00: Start transaction, read data as defined in Table A.82; When P1=02 and P2=01: read data as defined in Table A.83; When P1=03 and P2=00: Online data return When P1=03 and P2=01: online return of transfer transaction result
Lc	Data field data length
Data	See explanation of command message data fields
Le	00

Notes:

- 1) In IC card transfer mode, execute the “IC card transfer procedures” defined in Table A.61.
- 2) In card-not-present transfer mode, execute the “card-not-present channel transfer procedures” defined by Table A.61. In this mode, first utilize “primary account balance inspection transaction” to complete IC card verification and then complete “card-not-present channel transfer transaction”.

A.4.11.3 Command Message Data Fields**IC Card transfer procedures:****When P1=00 and P2=00:**

Start “IC card transfer procedures” ; see the command message data field table below:

Table A.74 Command Message Data Fields

Data	Length (in bytes)	Remarks
Transaction Amount	6	Transfer amount When this field is all 0 then utilize the terminal to input transfer amount; when not all 0, the terminal will confirm the transfer amount.
Transaction Date YYMMDD	3	Current date at processing center
Transaction Time HHMMSS	3	Current time at processing center
Transfer Target Account Number	20	Pad from right with spaces if insufficient Maximum card number length is currently 16 bytes with 1 byte being RFU
Transfer Target Cardholder Name	20	Numbers and Chinese characters are permitted; pad from left when insufficient.
Other Transaction Data	Variable (VAR)	See notes below table

Notes: Other transaction data is processing center data; during abnormal transactions (such as reversals, script notifications, etc.) this data must be sent back to the processing center in abnormal transaction messages (such as reversals and script notifications). Terminals do not need to analyze this data.

When P1=00 and P2=01:

The command message does not contain data fields.

When P1=01:

Online returned data; see the table below for command message data fields:

Table A.75 Command Message Data Fields

Data	Length (in bytes)	Remarks
Connection Result	1	00: Normal connection; 01: Unable to connect
Issuer Authorization Data (Tag 9F)	10-18	TLV format
Authorization Response Code (Tag 8A)	4	TLV format

Data	Length (in bytes)	Remarks
71 Script Data (Tag 71)	Variable	TLV format
72 Script Data (Tag 72)	Variable	TLV format

Card-not-present channel transfer procedures:

When P1=02 and P2=00:

Start “card-not-present channel transfer procedures” ; see the command message data field table below:

Table A.76 Command Message Data Fields

Data	Length (in bytes)	Remarks
Transaction Amount	6	Transfer amount When this field is all 0 then utilize the terminal to input transfer amount; when not all 0, the terminal will confirm the transfer amount.
Transaction Date YYMMDD	3	Current date at processing center
Transaction Time HHMMSS	3	Current time at processing center
Transfer Account Number	20	Pad from right with spaces if insufficient Maximum card number length is currently 16 bytes with 1 byte being RFU
Transfer Target Cardholder Name	20	Numbers and Chinese characters are permitted; pad from left when insufficient.
Other Transaction Data	Variable (VAR)	See notes below table

Notes:

- 1) Other transaction date refers to processing center data; during abnormal transactions (such as reversals, script notifications, etc.) this data must be sent back to the processing center in abnormal transaction messages(such as reversals and script notifications). Terminals do not need to analyze this data.

When P1=02 and P2=01:

The command message does not contain data fields.

When P1=03 and P2=00:

Online returned data; see the table below for command message data fields:

Table A.77 Command Message Data Fields

Data	Length (in bytes)	Remarks
Connection Result	1	00: Normal connection; 01: Unable to connect
Primary account Balance	6	BCD
Issuer Authorization Data (Tag 9F)	10-18	TLV format
Authorization Response Code (Tag 8A)	4	TLV format
71 Script Data (Tag 71)	Variable	TLV format
72 Script Data (Tag 72)	Variable	TLV format

Notes:

- 1) During “card-not-present channel transfer procedures”, “primary account balance inspection” is only used to verify user PIN and allow the terminal to receive the primary account balance; the primary account balance is not displayed.

When P1=03 and P2=01:

Online return of transfer results; see the table below for command message data fields:

Table A.78 Command Message Data Fields

Data	Length (in bytes)	Remarks
Connection Result	1	00: Normal connection; 01: Unable to connect
Authorization Response Code (Tag 8A)	4	TLV format Card-not-present channel transfer transaction authorization response code

A.4.11.4 Response Message Data Fields

IC Card transfer procedures:

When P1=00 and P2=00:

Start “IC card transfer procedures”; see the command message data field table below:

Table A.79 Response Message Data Fields

Data	Length (in bytes)	Remarks
Transaction result	1	01: Transaction declined; 02: Request online transaction
Transaction Amount	6	Transfer amount
Terminal Verification Result (TVR) (Tag 95)	7	TLV format
Transaction Date (Tag 9A)	5	TLV format
Random Number (Tag 9F37)	7	TLV format
Application Interchange Profile (AIP) (Tag 82)	4	TLV format
Application Transaction Code (ATC) Tag 9F36)	5	TLV format
Ciphertext Information Type (CID) (Tag 9F27)	4	TLV format
Application Ciphertext (AC) (Tag 9F26)	11	TLV format
Issuer Application Data (Tag 9F10)	Maximum 35	TLV format
Cardholder Verification Method (CVM) Result (Tag 9F34)	6	TLV format

Data	Length (in bytes)	Remarks
Transaction Sequence Counter (Tag 9F41)	5-7	TLV format
Special File Name (Tag 84)	7-18	TLV format
Application Version Number (Tag 9F09)	5	TLV format

Notes:

- 1) Terminal must return “transaction amount”; transfer amount is determined by this terminal response.

When P1=00 and P2=01:

Read PAN, PAN serial number, and related information: see the command response data field table below:

Table A.80 Response Message Data Fields

Data	Length (in bytes)	Remarks
PAN (Tag 5A)	Maximum 12	Application account number, TLV format
PAN Serial Number (Tag 5F34)	4	Application account number serial number, TLV format
Magstrip-Equivalent Data (Tag 57)	Maximum 21	Magstrip track 2 equivalent data, TLV format
Online PIN Ciphertext (Tag 99)	130	PIN/public key encrypted-online PIN ciphertext

When P1=01:

Returned online data sent from terminal to processing center; see the response message data field table below:

Table A.81 Load Online Response Message Data Fields

Data	Length (in bytes)	Remarks
Transaction result	1	00: Transaction approved; 01: Transaction declined
Script Execution Result (Tag DF31)	8	TLV format
Terminal Verification Result (TVR) (Tag 95)	7	TLV format
Application Transaction Code (ATC) (Tag 9F36)	5	TLV format
Ciphertext Information Type (CID) (Tag 9F27)	4	TLV format
Application Ciphertext (AC) (Tag 9F26)	11	TLV format
Issuer Application Data (Tag 9F10)	Maximum 35	TLV format

Cardless channel transfer procedures:**When P1=02 and P2=00:**

Start “card-not-present channel transfer procedures”; see the command message data field table below:

Table A.82 Response Message Data Fields

Data	Length (in bytes)	Remarks
Transaction result	1	01: Transaction declined; 02: Online Connection request
Terminal Verification Result (TVR) (Tag 95)	7	TLV format
Transaction Date (Tag 9A)	5	TLV format
Random Number (Tag 9F37)	7	TLV format

Data	Length (in bytes)	Remarks
Application Interchange Profile (AIP) (Tag 82)	4	TLV format
Application Transaction Code (ATC) Tag 9F36)	5	TLV format
Ciphertext Information Type (CID) (Tag 9F27)	4	TLV format
Application Ciphertext (AC) (Tag 9F26)	11	TLV format
Issuer Application Data (Tag 9F10)	Maximum 35	TLV format
Cardholder Verification Method (CVM) Result (Tag 9F34)	6	TLV format
Transaction Sequence Counter (Tag 9F41)	5-7	TLV format
Special File Name (Tag 84)	7-18	TLV format
Application Version Number (Tag 9F09)	5	TLV format

When P1=02 and P2=01:

Read PAN, PAN serial number, and related information: see the command response data field table below:

Table A.83 Response Message Data Fields

Data	Length (in bytes)	Remarks
PAN (Tag 5A)	Maximum 12	Application primary account number
PAN Serial Number (Tag 5F34)	4	Application primary account serial number
Magstrip-Equivalent Data (Tag 57)	Maximum 21	Magstrip-equivalent data

Data	Length (in bytes)	Remarks
Online PIN Ciphertext (Tag 99)	130	Online PIN ciphertext encrypted by PIN Key

When P1=03 and P2=00:

Returned online data sent from terminal to processing center; see the response message data field table below:

Table A.84 Response Message Data Fields

Data	Length (in bytes)	Remarks
Transaction result	1	00: Transaction approved; 01: Transaction declined
Transaction Amount	6	Transfer amount
Script Execution Result (Tag DF31)	8	TLV format
Terminal Verification Result (TVR) (Tag 95)	7	TLV format
Application Transaction Code (ATC) (Tag 9F36)	5	TLV format
Ciphertext Information Type (CID) (Tag 9F27)	4	TLV format
Application Ciphertext (AC) (Tag 9F26)	11	TLV format
Issuer Application Data (Tag 9F10)	Maximum 35	TLV format

- 1) Terminal must return “transaction amount” for successive card-not-present channel transfer transactions; transfer amount is determined by this terminal response. **When P1=03 and P2=01:**

No response data.

A.4.11.5 Response Message Status Codes

“9000” designates that this command was executed successfully.

Refer to Appendix A.5 – Terminal Command Response Status Code Table for possible error codes for this command.

A.5 Terminal Command Response Status Code Table

Table A.85 Terminal Command Response Status Code Table

SW1	SW2	Meaning
'90'	'00'	Normal processing
'61'	'XX'	Normal processing, 'XX' indicates the additional data length that can be retrieved using successive commands
'6C'	'XX'	Length error (Le inaccurate, 'XX' indicates actual length)
'65'	'81'	Terminal device error
'69'	'82'	Security status insufficient
'69'	'85'	Usage requirements insufficient
'69'	'86'	Command not permitted
'69'	'88'	Security message data object inaccurate
'67'	'00'	Data length error
'6A'	'80'	Data object does not exist
'6A'	'84'	Terminal memory capacity insufficient
'6A'	'86'	P1, P2 parameters inaccurate
'6D'	'00'	INS unsupported or error
'6E'	'00'	CLA unsupported or error

Appendix B

(Informative Appendix)

Basic Encryption Algorithms

B.1 RSA encryption algorithm

RSA public key is used to encrypt the plaintext message. Encryption standard uses RSAES-PKCS1-V1_5 in the encryption mode of PKCS # 1 specification.

- Algorithm parameter

M	Plaintext
mLen	Length of M
EM	Encoded plaintext message
C	Cryptogram message
K	Length of RSA public key modulus

- Steps of algorithm
 - Message encoding:
 - Generate a non-zero random byte string PS with the length of $k - mLen - 3$; PS has at least 8 bytes.
 - Connect PS and M in the following way, and generate a k-byte encoded message EM

$$EM = 0x00 \parallel 0x02 \parallel PS \parallel 0x00 \parallel M$$

- Encryption: Use RSA public key to encrypt EM and generate cryptogram C.

B.2 RSA signature algorithm

Use RSA private key to sign the message abstract. Signature standard uses RSASSA-PKCS1-V1_5 signature mode in the PKCS # 1 specifications. Message abstract uses SHA-1 algorithm.

- Algorithm parameter

M	Plaintext
EM	Encoded plaintext message
C	Cryptogram message

M	Plaintext
K	Length of RSA public key modulus

- Steps of algorithm

1. Calculate message abstract H of message M, where Hash algorithm uses SHA-1:

$$H = \text{SHA-1}(M)$$

2. Message encoding:

- a) Generate DER encoding T of H information, with the length of tLen:

$$T = (0x) 30 21 30 09 06 05 2b 0e 03 02 1a 05 00 04 14 \parallel H$$

- b) Generate a byte string PS, which is composed by $(k - tLen - 3) 0xFF$

- c) Connect PS and T by the following method, generating a k-byte encoding message EM

$$EM = 0x00 \parallel 0x01 \parallel PS \parallel 0x00 \parallel T$$

3. Signature:

Use RSA private key to sign on EM.

B.3 Symmetric encryption algorithm

In this specification, cipher block chaining (CBC) mode using the block cipher algorithm is used for data encryption and protection. Among which, the encryption algorithm can use AES algorithm with 128-bit key length or 3DES algorithm. Block length using AES encryption is 128 bits, and block length using 3DES encryption is 64 bits.

- Algorithm parameter

M	Plaintext
C	Cryptogram message
K	Encrypted key
IV	Initial vector
$E_K(M)$	Use key K to encrypt M
$D_K(C)$	Use Key K to decrypt C

- Steps of algorithm

AES block algorithm

1. Padding and grouping

Add 0x80 to the plaintext M, then pad a minimum 0x00 on the right, so after padding, length of message M ($M = (M \parallel 80 \parallel 00 \parallel 00 \parallel \dots \parallel 00)$) will be an integer multiple of 16. Divide M into 16-byte blocks, which are M1, M2, ..., Mn.

2. Encryption calculation

Use key K under CBC mode to encrypt the plaintext block M1, M2, ..., Mn to encryption block C1, C2, ..., Cn using AES-128 algorithm. Where, the initial vector $IV = (00 \parallel 00 \parallel 00 \parallel 00 \parallel 00 \parallel 00 \parallel 00 \parallel 00)$.

CBC mode encryption process is as follows:

$$C_0 = IV$$

$$C_i = E_K(M_i \oplus C_{i-1}), \quad i = 1, 2, \dots, n$$

$$C = (C_1 \parallel C_2 \parallel \dots \parallel C_n)$$

3. Decryption calculation

Use key K under CBC mode to decrypt the cryptogram into block C1, C2, ..., Cn to plaintext block M1, M2, ..., Mn. using AES-128 algorithm, Where, the initial vector $IV = (00 \parallel 00 \parallel 00 \parallel 00 \parallel 00 \parallel 00 \parallel 00 \parallel 00)$.

CBC mode decryption process is as follows:

$$C_0 = IV$$

$$M_i = D_K(C_i) \oplus C_{i-1}, \quad i = 1, 2, \dots, n$$

$$M = (M_1 \parallel M_2 \parallel \dots \parallel M_n)$$

3DES block algorithm

1. Padding and grouping

Add 0x80 to the plaintext M, then pad a minimum 0x00 on the right, so after padding, length of message M ($M = (M \parallel 80 \parallel 00 \parallel 00 \parallel \dots \parallel 00)$) will be an integer multiple of 8. M is divided into 8-byte blocks, which are M1, M2, ..., Mn

2. Encryption calculation

Use the key under CBC mode to encrypt the plaintext block M1, M2, ..., Mn to encryption block C1, C2, ..., Cn. using 3DES algorithm. Where, the initial vector $IV = (00 \parallel 00 \parallel 00 \parallel 00 \parallel 00 \parallel 00 \parallel 00 \parallel 00)$.

CBC mode encryption process is as follows:

$$C_0 = IV$$

$$C_i = E_K(M_i \oplus C_{i-1}), \quad i = 1, 2, \dots, n$$

$$C = (C_1 \parallel C_2 \parallel \dots \parallel C_n)$$

3. Decryption calculation

Use the key under CBC mode to decrypt the cryptogram block $C_1, C_2, \dots, C_n$ to plaintext block $M_1, M_2, \dots, M_n$ using 3DES algorithm. Where, the initial vector $IV = (00 \parallel 00 \parallel 00 \parallel 00 \parallel 00 \parallel 00 \parallel 00 \parallel 00)$.

CBC mode decryption process is as follows:

$$C_0 = IV$$

$$M_i = D_K(C_i) \oplus C_{i-1}, \quad i = 1, 2, \dots, n$$

$$M = (M_1 \parallel M_2 \parallel \dots \parallel M_n)$$

Appendix C

(Normative Appendix)

MAC Algorithm

C.1 Block algorithm based MAC

In accordance with ISO/IEC 9797-1 specification, MAC algorithm uses symmetric encryption algorithm with a key length of 128 bits under CBC mode to calculate the 8-byte MAC value for any length messages.

- Algorithm parameter

Table C.1 Description of MAC Algorithm Parameter

M	Plaintext
C	Cryptogram message
M _{AC}	Message authentication code
K	MAC key
IV	Initial vector
E _K (M)	Use key K to encrypt M
D _K (C)	Use Key K to decrypt C

- Steps of algorithm

Padding

Add 0x80 to the plaintext M, then pad a minimum 0x00 on the right, so after padding, length of message M ($M = (M || 80 || 00 || 00 || \dots || 00)$) will be an integer multiple of 8. Divide M into 16-byte blocks, which are M₁, M₂, ..., M_n.

MAC calculation

Use the left 8-byte KL of key K under CBC mode through 3DES algorithm to encrypt block M₁, M₂, ..., M_n. Where, the initial vector IV = (00 || 00 || 00 || 00 || 00 || 00 || 00 || 00).

CBC mode encryption process is as follows:

$$C_0 = IV$$

$$C_i = E_{KL}(M_i \oplus C_{i-1}), \quad i = 1, 2, \dots, n$$

The method to use the last block of to calculate the message authentication code is as follows:

$$M_{AC} = E_{KL}(D_{KR}(C_n))$$

C.2 HASH algorithm based on HMAC

- Algorithm parameter

According to FIPS specification, HMAC algorithm uses SHA-1 algorithm to generate HAMC.

Table C.2 Description of HMAC Algorithm Parameter

ipad	Pad the byte string, the content is: repeat padding 64 times of the 8-bit byte 0x36
opad	Pad the byte string, the content is: repeat padding 64 times of the 8-bit byte 0x5c
text	Input data whose MAC needs to be calculated, excluding the padded byte string
K	MAC key
t	Generated MAC byte length
Hash secure HASH algorithm	Use key K to encrypt M
$D_K(C)$	See FIPS 180-3

- Algorithm parameter

Use the following formula to calculate MAC value of data text:

$$MAC(text)_t = HMAC(K, text)_t = Hash((K_0 \oplus opad) || Hash((K_0 \oplus ipad) || text))$$

Detailed description is as follows:

- If $K = 64$, make $K_0 = K$. Go to Step 4;
- If $K > 64$, make $K_0 = Hash(K)$. Go to Step 4;
- If $K < 64$, pad 0x00 byte at the end of K, and generate 64-byte K_0 ;
- XOR operation of K_0 and ipad produces a 64-byte string: $K_0 \oplus ipad$;
- Add text to the end of string $K_0 \oplus ipad$ generated from Step 4: $(K_0 \oplus ipad) || text$;
- Apply Hash algorithm to string generated from step 5 and get: $Hash((K_0 \oplus ipad) || text)$;

7. XOR operation of K_0 and opad: $K_0 \oplus \text{opad}$;
8. Add the result generated from step 6 to the end of the result from step 7:

$$(K_0 \oplus \text{opad}) || \text{Hash}((K_0 \oplus \text{ipad}) || \text{text})$$

9. Apply Hash algorithm to the result of step 8 and get:

$$\text{Hash}((K_0 \oplus \text{opad}) || \text{Hash}((K_0 \oplus \text{ipad}) || \text{text})).$$

10. Hash value generated from Step 9 is taken as MAC value.

Appendix D

(Informative Appendix)

Certificate Format Table

Table 9 Certificate Format Table

Certificate Field Name	Meaning	Explanation		Segment Content (Example)
Version	Version number	Certificate version number		V3
Serial Number	Serial number	Determined by issuing institution		Certificate Serial Number
Signature	Signature Algorithm	In accordance with national standards		In accordance with national standards
Issuer	Issuer	C	Country	CN
		O	Unit	Issuer institution
Validity	Validity expiration			X months
notBefore	Start date of validity	Date of issuance		YYMMDD + HHMMSS
notAfter	End date of validity	Start date + x months		YYMMDD + HHMMSS
Subject	Subject	C	Country	CN
		O	Institution	Issuing institution
		OU	Institution name	XX bank or XX institution
		OU	Certificate Type	Includes all 4 integer types
		CN	General name	Device name, domain name or IP, manufacturer name etc.

Certificate Field Name	Meaning	Explanation	Segment Content (Example)
Subject Public Key Information	Public Key	Includes encryption algorithm and public key value	Uses RSA algorithm; key length greater or equal to 1024
Extensions	Extension field		
KeyUsage	Key usage method	Key extension	Digital Signature,
			keyEncipherment;
			Other required algorithms
extKeyUsage	Extension key usage		If key usage method is limited to indicate usage only then the label is crucial; otherwise the label is not.
basicConstraints	Basic constraints		CA certificate label is crucial, actual terminal certificate label is not crucial
SubjectKeyIdentifier	Subject Key Identifier	Key extension	Hash value of user certificate public key
AuthorityKeyIdentifier	Authority Key Identifier	Key extension	Authority certificate public key hash value
CRLDistributionPoints	CRL Distribution Points	Key extension	[1]CRL Distribution Point Distribution Point Name: Full Name: Directory Address: DN...
			[2]CRL Distribution Point Distribution Point Name: Full Name: URL=http://ldap.xxca.org.cn/crl/caxcrlxx.crl

Certificate Field Name	Meaning	Explanation	Segment Content (Example)
SignatureAlgorithm	Signature Algorithm	Algorithm used to electronically sign certificate basic information	
Issuer's Signature	Issuer Signature	Issuer electronic signature of certificate basic information	Electronic signature value

Appendix E

(Informative Appendix)

Secure Channel Establishment Procedure Example

The host sends a GET CLIENT HELLO command to the terminal; terminal responds with a 1-byte algorithm identifier and 32-byte random number;

The host uses the 1-byte algorithm identifier sent by the terminal to acquire the list of supported symmetrical and assymetrical algorithms; this identifier is transmitted to the processing center which uses the list to determine if the signature algorithm and symmetrical algorithm used by the processing center trusted server certificate is supported;

The random number is used after this point for verification purposes and generation of authentication information.

The server then sends the trusted server certificate, algorithm identifier, and 32-byte random number to the terminal.

At this point, the terminal should first inspect the legality of the trusted server certificate using HASH SERVER CERTIFICATE and VERIFY SERVER CERTIFICATE commands;

The HASH SERVER CERTIFICATE command is used to conduct a hash operation on the main body of the trusted server certificate and then save the result of the hash operation within the terminal;

The VERIFY SERVER CERTIFICATE command use Root CA certificate to decrypts the trusted server certificate and then compares it to the hash result of the HASH SERVER CERTIFICATE command; this determines the legality of the trusted server certificate;

Once the legality of the trusted server certificate has been determined terminal will generate a 48-byte random number as a shared main key and then encrypt this key using the trusted server certificate; the EXPORT MASTERKEY command will then complete the encryption of the shared main key by the trusted server certificate and return a 128-byte ciphertext;

The terminal certificate will then be read using the READ CERTIFICATE and GET CERT RESPONSE commands; because certificates are larger than the maximum transmission bytes admissible by CCID, making it impossible to read a terminal certificate using a single command. The GET CERT RESPONSE command can be used multiple times until the entire terminal certificate has been read; for command details, refer to A.3;

To authenticate the terminal the processing center must verify the terminal private key signature. The terminal will use the CLIENT SIGN command to conduct a hash operation and signature of the connection value of the random number inputed

from the terminal and the terminal's own random number, then return the 128-byte signature data;

The terminal then sends the 128-byte shared main key ciphertext, terminal certificate, and 128-byte signature value to the processing center.

This server uses the root certificate to verify the legality of the terminal certificate; if successful then it will verify the signature value with the terminal public key and thus verify the legality of the terminal itself; once the terminal has been authenticated the trusted server certificate private key will be used to decrypt the shared main key ciphertext and acquire the 48-byte shared main key; here, a server authentication completion message must be sent; so as to guard against this message being fraudulent it must be completed using an HMAC calculation; the first 16 values of the 48-byte shared main key serves as the key with the data consisting of ASCII"SERVER", the terminal random number, the processing center random number, the trusted server certificate hash value, the terminal certificate hash value, the signature value sent from terminal to the server, and the shared main key ciphertext;

The processing center sends the handshake completion information—the HMAC value calculated by the processing center—and sends it to the terminal.

Once the terminal has received the HMAC it will use the HMAC (P2=0x01) command to verify the HMAC value produced by the server during handshake completion. Then the HMAC (P2=0x00) command will be used to return the HMAC value to the terminal; this process is the same as used by the processing center to generate an HMAC value, but changes ASCII"SERVER" to "CLIENT";

The processing center sends the handshake completion information—the HMAC value calculated by the processing center—and sends it to the terminal;

The terminal will then use the HMAC (P2=0x02) command to generate a dialogue key; this dialogue key will only be stored within the terminal and will not be exported; in the case of a power failure this key must be regenerated.

The processing center then verifies the terminal handshake completion information and generates a session key.

Finally, both the processing center and the terminal will have a 48-byte shared main key and a 20-byte session key; the first 16 bytes of the 20-byte session key acts as the encryption key while the last 16-bytes act as the key used to calculate MAC.

Appendix F

(Normative Reference)

Requirements for Terminal Support of Dual Processing Centers

The terminal should be capable of processing transactions with two different processing center systems. Terminals can determine the unique identity of a processing center using the distinguished name (DN) of that processing center's trusted server certificate.

F.1 Terminal Processing Procedures

When a terminal connects to a processing center and conducts an online transaction, the terminal must determine the identity of the connected processing center and conduct all necessary terminal functions; this includes terminal transaction application processes and choosing of appropriate channel PIN encryption certificates and encrypted PIN data.

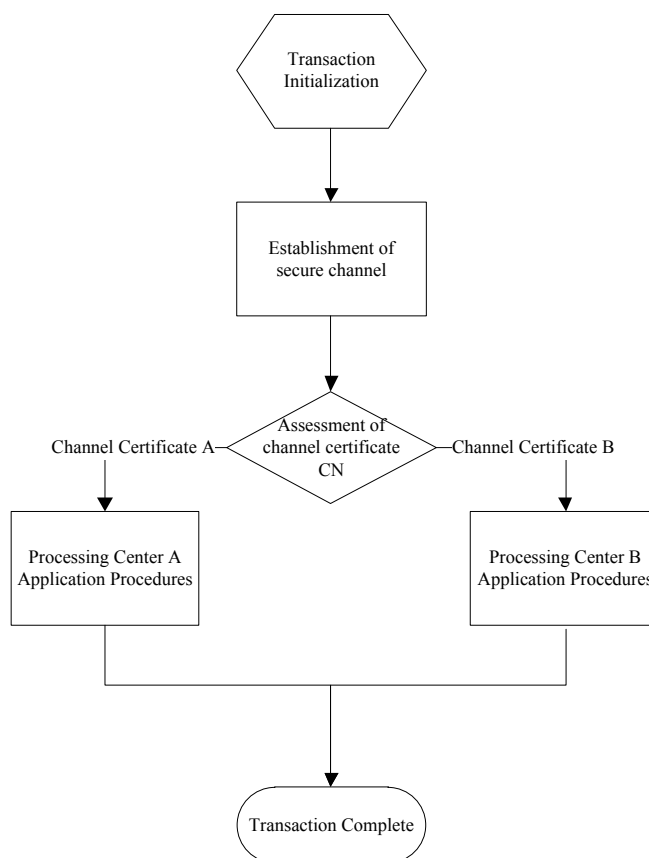


Chart F.1 - Terminal Processing Procedures

- 1) Cardholder begins online transaction;
- 2) Processing center and terminal establish secure channel;
- 3) The terminal uses the trusted server certificate general name content of the DN field retrieved during the establishment of the secure channel to verify

the identity of the processing center. If the trusted server certificate is for processing center A, then the terminal will enter into processing center A terminal application procedures; if the trusted server certificate is for processing center B, then the terminal will enter into processing center B terminal application procedures;

- 4) The terminal will then execute processing center terminal applications in accordance with step 3 (including usage of PIN encryption certificates, reversal mechanisms, display information prompts, and all other mechanisms appropriate for that processing center);
- 5) Transaction procedures are completed and the transaction ends.

F.2 DN Rules

See Table D. 1 for DN characteristics and meanings.

Table F.1 DN Characteristics

DN Characteristics	Meaning	Remarks
C	Company Country Name	C = CN
O	Certificate Issuing Institution	Example: CFCA
OU	Certificate Application Institution	Terminal Device Manufacturer
OU	Certificate Type	Example: UP Reader
CN	Certificate Holder (Terminal Device) General Name	1、 Standard name main entity of entity institution certificate 2、 Certificate main device domain name or IP of server certificate 3、 Terminal device code of terminal certificate

F.3 Terminal Personalization Differences

The terminal data and certificate system of a terminal that supports dual processing centers are the same as those used for terminals that only support a single processing center, but the number of certificates used for terminal personalization is very different. See Table D.1 for personalization certificate amounts.

Table F.2 Personalization Certificate Amounts

Terminal Capabilities Certificate Type	Single Processing Center Terminal	Dual Processing Center Terminal
Terminal Certificate	1	1
Root CA Certificate	1	1
PIN Encryption Certificate	1	2
Trusted server certificates	Personalization not required	Personalization not required

Note: Trusted server certificates established during the establishment of a secure channel are sent to the terminal by the processing center for verification.

Appendix G

(Normative Appendix)

List of Command Status Code

Table 10 Terminal Command- List of Return Code

SW1	SW2	Default Meaning of Status Code	READ TERMINAL INFO	MANAGE BUZZER	MANAGE LED	CONFIG DISPLAY FORMAT	ADD CERTIFICATE	UPDATE CERTIFICATE	DELETE CERTIFICATE	ENCRYPT ISO COMMAND	MANAGE SECURE CHANNEL	READ NON-PBOC PIN	UPDATE TERMINAL PIN	CREDIT FOR LOAD	DEBIT FOR PURCHASE	GET ELECTRONIC CASH BALANCE	GET PRIMARY BALANCE	SET APP PARAMETERS	GET PBOC TAG VALUE
61	XX	Normal processing																	
62	00	No information provided																	
62	81	Returned data may be wrong																	
62	82	Length of file < Le																	
62	83	Selected file invalid							√	√	√	√	√	√	√	√	√		
62	84	FCI format does not match that specified by P2																	
63	00	Authentication failure							√	√	√	√	√	√	√	√	√		
63	CX	Verification fails, X attempts remaining																	
64	00	Status mark bit is unchanged																	
65	81	Memory error	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√

SW1	SW2	Default Meaning of Status Code	READ TERMINAL INFO	MANAGE BUZZER	MANAGE LED	CONFIG DISPLAY FORMAT	ADD CERTIFICATE	UPDATE CERTIFICATE	DELETE CERTIFICATE	ENCRYPT ISO COMMAND	MANAGE SECURE CHANNEL	READ NON-PBOC PIN	UPDATE TERMINAL PIN	CREDIT FOR LOAD	DEBIT FOR PURCHASE	GET ELECTRONIC CASH BALANCE	GET PRIMARY BALANCE	SET APP PARAMETERS	GET PBOC TAG VALUE
67	00	Length error	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√
68	82	Do not support secure message																	
69	00	Unable to process	√	√	√		√		√	√	√	√	√	√	√	√	√	√	√
69	01	Command unaccepted (invalid)																	
69	81	Command incompatible with file structure																	
69	82	Security status is not met							√	√	√	√	√	√	√	√	√	√	√
69	83	Verification method lockup							√	√	√	√	√	√	√	√	√	√	√
69	84	Reference data invalid																	
69	85	The application conditions are not met	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√
69	86	Does not meet the conditions for command execution (non-current EF)																	
69	87	Security message data missing							√	√	√	√	√	√	√	√	√	√	√

SW1	SW2	Default Meaning of Status Code	READ TERMINAL INFO	MANAGE BUZZER	MANAGE LED	CONFIG DISPLAY FORMAT	ADD CERTIFICATE	UPDATE CERTIFICATE	DELETE CERTIFICATE	ENCRYPT ISO COMMAND	MANAGE SECURE CHANNEL	READ NON-PBOC PIN	UPDATE TERMINAL PIN	CREDIT FOR LOAD	DEBIT FOR PURCHASE	GET ELECTRONIC CASH BALANCE	GET PRIMARY BALANCE	SET APP PARAMETERS	GET PBOC TAG VALUE
69	88	Data objects of security information is incorrect																	
6A	80	Data field parameter is incorrect																	
6A	81	Functions not support							√	√	√	√	√	√	√	√	√	√	√
6A	82	File not found																	
6A	83	No records found																	
6A	84	No enough storage space in file																	
6A	86	P1, P2 parameter is incorrect	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√
6A	88	Reference data not found																	
6B	00	Parameter error (offset address outside the EF)																	
6C	XX	Length error (Le error; 'XX' is the actual length)																	
6F	00	Invalid data	√	√	√		√		√	√	√	√	√	√	√	√	√	√	√

SW1	SW2	Default Meaning of Status Code	READ TERMINAL INFO	MANAGE BUZZER	MANAGE LED	CONFIG DISPLAY FORMAT	ADD CERTIFICATE	UPDATE CERTIFICATE	DELETE CERTIFICATE	ENCRYPT ISO COMMAND	MANAGE SECURE CHANNEL	READ NON-PBOC PIN	UPDATE TERMINAL PIN	CREDIT FOR LOAD	DEBIT FOR PURCHASE	GET ELECTRONIC CASH BALANCE	GET PRIMARY BALANCE	SET APP PARAMETERS	GET PBOC TAG VALUE
6F	01	PIN code not entered										√	√	√	√	√	√	√	√
90	00	Successful execution, no error	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√
93	01	Insufficient amount																	
93	02	Insufficient amount																	
93	03	Application permanently locked							√	√	√	√	√	√	√	√	√	√	√
94	01	Insufficient amount														√			
94	02	Transaction counter reaches the maximum value																	
94	03	Key index not support	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√
94	06	Required MAC is not usable																	
6E	00	Types not supported: CLA error	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√

SW1	SW2	Default Meaning of Status Code	READ TERMINAL INFO	MANAGE BUZZER	MANAGE LED	CONFIG DISPLAY FORMAT	ADD CERTIFICATE	UPDATE CERTIFICATE	DELETE CERTIFICATE	ENCRYPT ISO COMMAND	MANAGE SECURE CHANNEL	READ NON-PBOC PIN	UPDATE TERMINAL PIN	CREDIT FOR LOAD	DEBIT FOR PURCHASE	GET ELECTRONIC CASH BALANCE	GET PRIMARY BALANCE	SET APP PARAMETERS	GET PBOC TAG VALUE
6D	00	Instruction code not supported	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√
66	00	Receiving response time out																	
66	01	Parity error of received character																	
66	02	Checksum is incorrect																	
66	03	Current DF file has no FCI																	
66	04	No SF or KF under current DF																	

Appendix H

(Normative Appendix)

Terminal Version Number Definition And Upgrade Rules

H.1 Version Number Definition

Terminal firmware version number is formatted as follow:

A.B.C.D

And, A\B\C\D are all decimal based, range 0-99, and code is ASCII character, e.g.: '1.1.1.1'.

H.2 Upgrade Rule

If terminal firmware version is Ac, Bc, Dc, and firmware version on firmware upgrade platform is As.Bs.Cs.Ds, then upgrade rules should be:

- 1) If $A_s > A_c$, then upgrade, if $A_s < A_c$, then do not upgrade, if $A_s = A_c$, go to step 2)
- 2) If $B_s > B_c$, then upgrade, if $A_s < A_c$, then do not upgrade, if $B_s = B_c$, go to step 3)
- 3) If $C_s > C_c$, then upgrade, if $A_s < A_c$, then do not upgrade, if $C_s = C_c$, go to step 4)
- 4) If $D_s > D_c$, then upgrade, if $A_s < A_c$, then do not upgrade.

Notes: 1) all comparison above are based on number, but not string; e.g.: if based on string comparison '9' is greater than '10', but if based on number comparison, 10 is greater than 9, and this specification is based on number comparison.

Appendix I

(Normative Appendix)

Certificate Update Flow

I.1 CA Certificate Update Flow

After secure channel established, MiniPay transaction processing system will use terminal CA certificate serial number to decide whether update or not, detailed update flow are defined as follow:

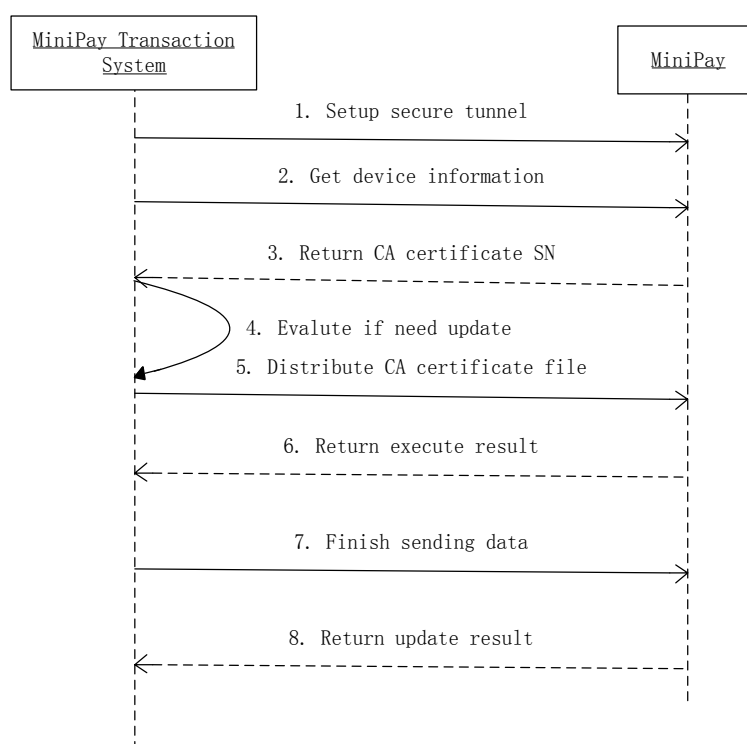


Figure I.1 CA certificate update flow

Steps:

- 1) After secure channel established, before online transaction starts, MiniPay system sends command 7E10 to get terminal information;
- 2) MiniPay system gets CA certificate serial number from terminal information;
- 3) MiniPay system compares CA certificate serial number, and evaluates certificate validation date;
- 4) If CA certificate already expired, or out of supported range, then new CA certificate will be distributed to terminal.

I.2 PIN Encryption Certificate Update Flow

After secure channel established, MiniPay system will use terminal PIN public key certificate serial number to decide if need update, detailed flow as below:

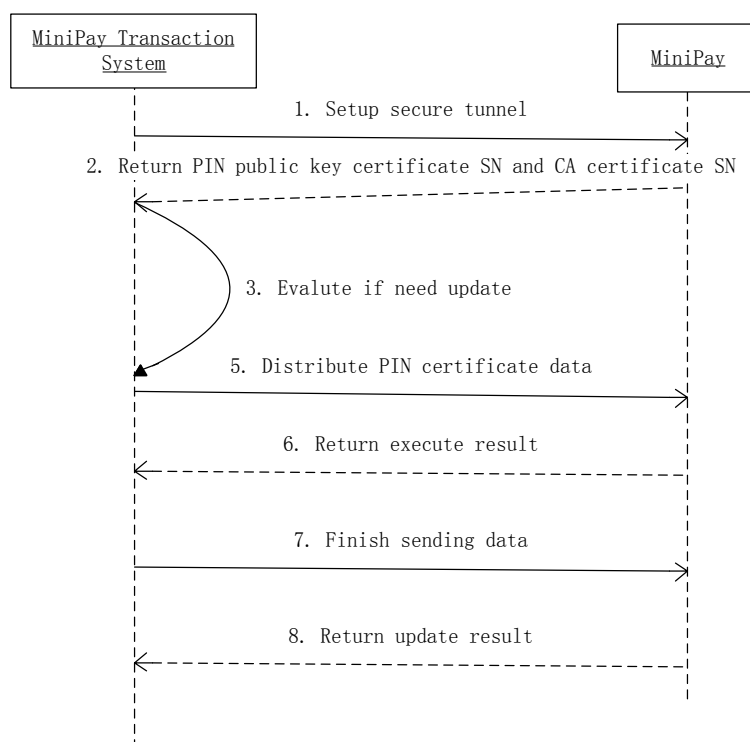


Figure I.2 PIN certificate update flow

Steps:

- 1) After secure channel established, before online transaction starts, MiniPay system send command 7E10 to get terminal information;
- 2) MiniPay system gets PIN certificate serial number and CA certificate serial number from terminal information;
- 3) MiniPay system uses PIN certificate serial number and CA certificate serial number to evaluate if terminal needs to update PIN public key and will choose update PIN certificate file, CA certificate and PIN certificate which match each other. MiniPay system uses CA certificate serial number to choose corresponding PIN public key certificate file;
- 4) MiniPay system compares selected PIN public key certificate file serial number and PIN certificate serial number returned by terminal, it will distribute new PIN public key certificate if they are different, or continue if they are the same;
- 5) After data is received, terminal uses CA certificate to validate new PIN public key certificate. Terminal will remove old certificate file if the validation

succeeds, and new PIN certificate encryption will be used in the following transactions.

I.3 Terminal Certificate Update Flow

After secure channel established, MiniPay system uses terminal returned terminal certificate validation date to decide if terminal certificate needs update. When update terminal certificate, MiniPay system will check if CA root certificate needs update. If yes, the system will update CA root certificate, then update terminal certificate. CA root certificate update and terminal certificate update need happen at the same time.

The detailed processing flows refer to below diagram:

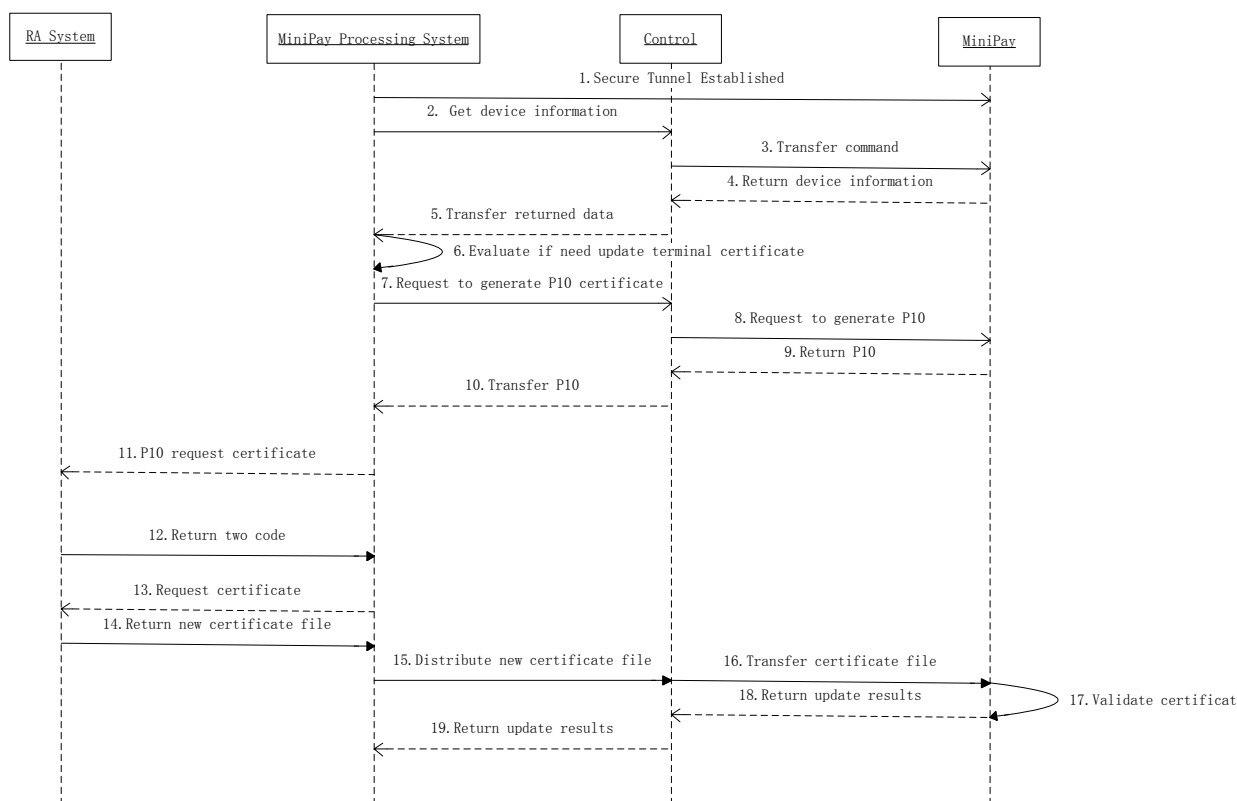


Figure I.3 Terminal certificate update flow

Steps:

- 1) After secure channel established, MiniPay processing system requests terminal to update terminal certificate

- 2) Terminal generates new public/private key pairs, format PKCS#10 certificate request file
- 3) Terminal returns PKCS#10 file
- 4) MiniPay processing system links RA system to request two code
- 5) RA system returns two code
- 6) MiniPay processing system uses two code, downloads public key certificate
- 7) RA system returns new public key certificate file
- 8) Server returns new public key certificate file, and use scontrol to transfer to terminal
- 9) Terminal validates new public key certificate signature, if success then deletes old terminal certificate file and private key file, complete update. During terminal update certificate process, if update failed, then terminal will save old terminal certificate and private key, until update success, then delete old terminal certificate and private key file.

I.4 Channel Certificate Configuration Requirement

In order to adapt terminal devices with different CA certificate length, SecurePay gateway server will use '7E25' command to identify the certificate type supported by terminal, then choose proper length channel certificate to communication. Detailed process refers to below diagram:

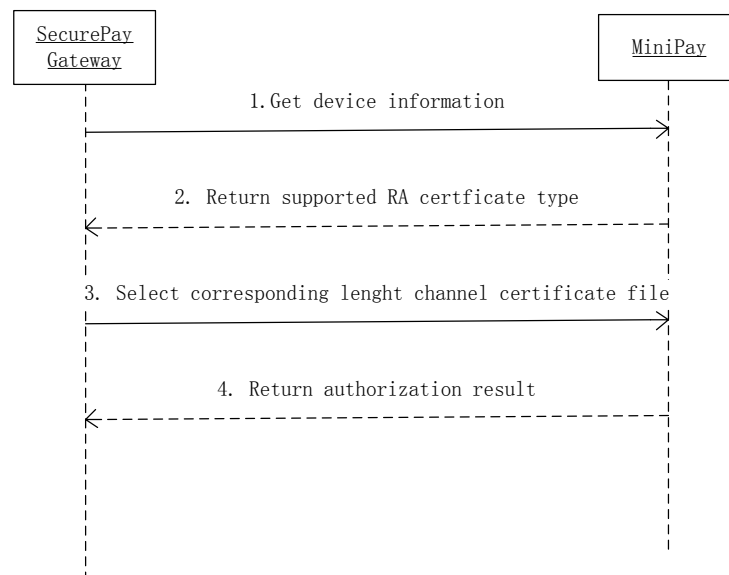


Figure I.4 Channel certificate self-adjust processing flow

Steps:

- 1) SecurePay gateway uses 7E25 command to get the CA certificate type supported by terminal
- 2) SecurePay gateway checks terminal CA certificate type, and chooses corresponding channel certificate to authenticate the device at the server side.
- 3) The Device end configures a suite of CA certificate, which is used to authenticate the channel certificate, in order to accomplish dual direction authentication between channel server and terminal certificate.

Appendix J
(Informative Appendix)
Transaction Type

Table J.1 Transaction Type Defination

Value	Transaction Type
0x00	Online Purchase
0x60	Recharge
0x31	PAN balance inquiry
0x40	Remittance
0x47	Credit card repayment

Appendix K

(Informative Appendix)

Platform Access And Communicate Interface Code

Table K.1. Platform Access Code

Platform Access (7E25 P1)	7E25 P1 code
windows	00
mac OS	01
windows	10
windows Pad	11
mac OS	12
iOS iPad	13
iOS iPhone	14
Android Pad	15
Android Phone	16
Set Top Box	02
Others	03(Linux)

Table K.2. Communication Interface Code

Transaction Communication Interface Type	Interface Type Code
USB	00
Acoustic	01
Bluetooth	02
APPLET	03
NFC	04
Set Top Box	05
Others	06